Universal Rank-Three Obstructions in Binary Arithmetic Occurrence Geometry

Circuit Decomposition Rigidity, Vector-Space Partitions, and Universal Non-Embedding Theorems

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Abstract

Let M be a binary matrix with columns $c_1, \dots, c_n$ and $C = \ker M \subseteq \mathbb{F}_2^n$ its code. The nonzero codewords, read as $0/1$ vectors, are the integer points of the box $[0,1]^n$ in the lattice $\Lambda_M = \{\alpha \in \mathbb{Z}^n : M\alpha \equiv 0 \bmod 2\}$. Conformally minimal (Graver) lattice vectors with corners in this fibre span parallelepipeds, the *cells*. The box is *embedded* if the cells meet face to face and *non-embedded* otherwise; a non-embedded box contains a *collision* of two cells with intersecting relative interiors, and its *minimum witness rank* $r_{\min}$ is the least dimension of the affine span of a collision.

We prove, from the definitions and without recourse to any computation in the central steps: (i) a box is non-embedded if and only if some codeword has two distinct decompositions into disjoint circuits (minimal supports); (ii) in a coordinatewise-minimal non-embedded box the all-ones word lies in C, all other words have a unique circuit decomposition, and the reduced coordinate set is a spanning subset X of an affine space $\mathrm{AG}(k-1,2)$ with $|X| = p \geq k+1$; (iii) the partitions of the all-ones word into circuits give a vector-space partition of $C/\langle \mathbf{1} \rangle \cong \mathbb{F}_2^{k-1}$, which satisfies an exact counting identity and a Parseval identity; (iv) these force at least $p-1$ two-circuit partitions of $\mathbf{1}$ when $p \geq 9$, sharp for $X = \mathrm{AG}(k-1,2)$; and, with a finite check for $p \leq 8$, *every* minimal non-embedded box has at least two such partitions; (v) consequently $r_{\min} = 3$ for *every* non-embedded box of the frozen binary model, and there is no minimal box with $r_{\min} \geq 4$. We also give an explicit rank-three obstruction (the even-weight code of length 4), the complete classification of minimal reduced boxes for nullity $3, 4, 5$ (with $1, 3, 12$ classes; reproduced here by an independent exhaustive enumeration) and, as source-reported results with independently re-checked representatives, nullity 6 (215 classes).

The scope is the frozen binary $0/1$ Graver-cell model. The results do not assert that every binary code is realized by a rational arithmetic occurrence fibre, and rank three is a statement about a local witness: it is not a polynomial-time algorithm for deciding embeddedness. Proof status, stated precisely: for $p \geq 9$ reduced coordinates the universal statements (including $n_2 \geq p-1$) are proved analytically in the text; the case $p \leq 8$ is closed by a complete, exhaustive finite enumeration (a computer-assisted step, with own code and exact arithmetic), so the universality of W2 and hence $r_{\min} = 3$ rest on that enumeration together with the analytic argument. Classification counts for nullity $3, 4, 5$ are re-enumerated here; the nullity-6 list of 215 classes is source-reported, its completeness is inherited from the release and no universal claim depends on it.

Keywords: binary linear codes, circuits, Graver basis, conformal minimality, vector-space partitions, finite affine geometry, non-embedding, Parseval identity.

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Part I
Foundations

1 Introduction: the geometry of non-embedding

1.1 The question

A binary matrix $M \in \mathbb{F}_2^{m\times n}$ defines a lattice

$$\Lambda_M = \{\alpha \in \mathbb{Z}^n : \ M\alpha \equiv 0 \pmod 2\},$$

and the integer points of the unit box $[0,1]^n$ in Λ_M, with the origin deleted, form a finite set F, the *fibre*. Identifying a $0/1$ vector with a word of $\mathbb{F}_2^n$, the fibre is the set of nonzero words of the binary code $C = \ker M$. Inside this lattice there is a canonical family of lattice vectors, the *Graver basis*: the conformally minimal nonzero elements of Λ_M. A *cell* is a parallelepiped spanned by linearly independent Graver vectors all of whose corners lie in F. The union of all cells is a geometric object assembled from the fibre in a way that depends on nothing but the lattice.

The question of this paper is the following. *When do the cells of the box fit together as a polyhedral complex, meeting only in common faces?* We call the box *embedded* when they do, and *non-embedded* when two distinct cells have intersecting relative interiors. In the latter case the pair is a *collision*. The smallest affine space containing a collision is its *witness space*, and the least dimension of such a space is the *minimum witness rank* $r_{\min}$ of the box.

The motivation comes from the arithmetic of occurrences in the UNNS nest recursion, where relation lattices of precisely this congruence type arise and where the cells describe which exponent vectors are connected by minimal moves [20, 21, 22, 23]. Nothing below depends on the arithmetic origin. The frozen model used here is purely combinatorial: it consists of the lattice Λ_M, its Graver basis, the box $[0,1]^n$, and the cells defined above. All objects are defined in Sections 2–4, and every statement is proved from those definitions.

1.2 A first picture

The smallest non-embedded box has $n = 4$ and $C = E_4$, the even-weight code of length 4 (a single parity check $M = (1\,1\,1\,1)$). Its fibre is the seven nonzero even words: the six words of weight two and the word 1111. The six weight-two words are the vertices of a regular octahedron in the affine hyperplane $\sum x_i = 2$ of $\mathbb{R}^4$. Three of the cells are the three *equatorial squares* of that octahedron, spanned by pairs of disjoint two-element supports: $\{12,34\}$, $\{13,24\}$, $\{14,23\}$. Any two of the three squares share a diagonal and all three contain the centre $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ in their relative interiors. The three squares do not meet in common faces, so the box is non-embedded, and any two of them span only a 3-dimensional affine space; see Figure 6 and Section 18.

The purpose of this paper is to show that this picture is universal. Every non-embedded box of the frozen binary model, whatever its length n and nullity $k = \dim C$, contains a collision that is as small as that one: two cells spanning an affine space of dimension *exactly three*, and none spans fewer. The proof goes through a rigid combinatorial structure on the circuits of minimal non-embedded boxes: a vector-space partition of $\mathbb{F}_2^{k-1}$ and two quadratic identities that force many two-circuit partitions of the all-ones word.

1.3 Main results

Throughout, a *box* is a binary code $C \leq \mathbb{F}_2^n$ (Definition 2.1); circuits are minimal supports of nonzero words; a box is *minimal* if it is non-embedded but every box obtained by deleting one coordinate (shortening) is embedded. The following statements are proved in this paper. The numbering of the corresponding source theorems in the GEO-I series is given in Appendix M.

Theorem A (circuit-partition criterion; Theorem 5.2). A box is non-embedded if and only if some set of coordinates has two distinct partitions into circuits; equivalently, some codeword has two distinct decompositions into disjoint circuits.

Theorem B (structure of minimal boxes; Theorems 7.1, 8.2 and 10.1). In a minimal non-embedded box the all-ones word $\mathbf{1}$ lies in the code, no circuit is a single coordinate, every codeword other than 0 and $\mathbf{1}$ has a *unique* decomposition into disjoint circuits, and, after identifying coordinates that agree on every codeword, the box is a spanning subset X of the affine space $\mathrm{AG}(k-1,2)$ with $|X| = p \geq k+1$. Isomorphism classes correspond to $\mathrm{AGL}(k-1,2)$-orbits of such subsets.

Theorem C (vector-space partition; Theorems 11.3, 12.1 and 13.3). The partitions P of $\mathbf{1}$ into circuits, with m_P blocks, give subspaces $V_P \leq C$ of dimension m_P whose images in $C/\langle\mathbf{1}\rangle \cong \mathbb{F}_2^{k-1}$ partition the nonzero vectors, so that

$$\sum_P \bigl(2^{m_P} - 2\bigr) = 2^k - 2, \qquad \sum_P \bigl(2^{m_P} S_P - 2p^2\bigr) = p\,(2^k - 2p),$$

where S_P is the sum of the squared block sizes.

Theorem D (many two-block partitions; Theorems 15.1 and 16.3). If $p \geq 9$ the number n_2 of partitions of $\mathbf{1}$ into exactly two circuits satisfies $n_2 \geq p-1$, with equality for $p \geq 10$ only for $X = \mathrm{AG}(k-1,2)$. For $p \leq 8$ a finite check shows $n_2 \geq 3$. Hence every minimal non-embedded box has $n_2 \geq 2$ (condition W2).

Theorem E (universal rank three; Theorem 17.1). Every non-embedded box of the frozen binary model has $r_{\min} = 3$. In particular no minimal non-embedded box has minimum witness rank 4 or more.

Section 18 gives the explicit rank-three obstruction for E_4 together with infinite families (Reed–Muller supports $\mathrm{AG}(k-1,2)$), and Section 19 reports the exact classification of minimal reduced boxes for $k = 3,4,5,6$ (1, 3, 12 and 215 classes). The classification is *not* used in the proof of Theorem E.

1.4 Proof architecture

The logical dependencies are summarised in Figure 13. The route is:

Thm A $\Rightarrow$ Thm B (full support, rigidity, affine reduction) $\Rightarrow$ Thm C $\Rightarrow$ Thm D $\Rightarrow$ Thm E.

The last step is the combination of three ingredients: a local construction showing that two distinct two-circuit decompositions of one word produce a collision of rank three; a general lower bound, valid for *every* collision, showing rank three is the least possible; and a reduction showing that every non-embedded box contains a minimal one by shortening, with collisions lifting to the larger box without changing their rank.

1.5 Scope and non-claims

1. *Frozen model.* All results concern the frozen binary 0/1 Graver-cell model defined in Section 2: the box $[0,1]^n$ of the lattice Λ_M over $\mathbb{F}_2$. Other moduli, alphabets, larger boxes $Q > \mathbf{1}$ or non-binary lattices are not treated (Section 23).
2. *Realizability.* The model contains a cell structure for *every* binary code C. The paper does **not** claim that every binary code arises from a rational arithmetic occurrence fibre, nor that any particular code does. The relation of the model to arithmetic is discussed only in the companion manuscript [20] and the cited releases.
3. *Meaning of $r_{\min}$.* The number $r_{\min}$ is the *minimum* over collisions of the dimension of the affine span of the collision. It is not the ambient dimension n, not the nullity $k = \dim C$, and not the rank of every witness: individual collisions can have large rank (Remark 4.9).
4. *No algorithmic claim.* That a rank-three witness always exists does not give a polynomial-time algorithm for deciding embeddedness. Section 20 discusses the difference between the size of the *local* witness and the *global* difficulty of locating it.
5. *Independence from Paper A.* The present paper can be read without the companion manuscript [20]. The two papers share the underlying GEO-I releases but treat different problems (the planar modular lattice $\Gamma_{d,s}$ there; binary relation lattices of arbitrary length here). Notation differences are reconciled in Appendix A; in particular the number of prime coordinates is written k in [20] and n here, while k denotes the nullity $\dim C$ in this paper.

1.6 Provenance and what the paper contributes

The mathematics presented here was developed and recorded in the archived research releases GEO-I v006–v011 [26, 27, 28, 29, 30, 31] (and checked against the later release [32]), all cited at the single commit `343f9aef2b6697466058165c5fc2ddf0f00e882b` of the repository. Those releases are version-stamped research records; they are *not* peer reviewed. This manuscript re-derives the proofs from the definitions so that no step depends on reading them. The relation to the source theorems is stated in each result (as a *Source status* line) and collected in the crosswalk of Appendix M. Where this manuscript adds material that we did not find in the release files we read, it is flagged in Appendix M (items marked "new here"); these include a finite check for $p \leq 8$ that replaces the use of the nullity-6 enumeration in the proof of W2, a closed formula for the number of cells through the centre of the box (Proposition 5.4), and the equality analysis in Theorem D.

Three kinds of evidence are kept apart throughout: (a) *universal proofs* in the text; (b) *finite enumerations* we re-ran with code written for this manuscript (Appendix L); (c) *source-reported* results, in particular the completeness of the nullity-6 classification, which we did not re-derive. Finite checks are never used as proofs of universal statements. The only finite computations on which a proof in the text depends are the enumeration for $p \leq 8$ in Section 16 (which is a complete and small enumeration of explicit objects, described exactly in Appendix K) and the verification of numerical identities in examples.

1.7 Organization

Part I (Sections 1–4) sets up codes, circuits, Graver cells, collisions and witness rank. Part II (Sections 5–9) proves the circuit-partition criterion, the structure of minimal boxes and the local rank-three extraction. Part III (Sections 10–17) is the heart of the paper: vector-space

partitions, counting and Parseval identities, inequalities, and the universality theorems. Part IV (Sections 18–20) gives explicit obstructions, the finite classification and consequences. Part V (Sections 21–23) compares the results to established mathematics, documents provenance and reproducibility, and lists open questions. Appendices A–M contain notation, auxiliary proofs and derivations, tables, and the theorem crosswalk.

2 Binary relation lattices and occurrence coordinates

2.1 Codes, supports, words

All vector spaces are over $\mathbb{F}_2$. We write $[n] = \{1, \dots, n\}$ and identify a word $w \in \mathbb{F}_2^n$ with its support $\operatorname{supp} w = \{i : w_i = 1\}$; sums of words are symmetric differences of supports. The weight is $\operatorname{wt}(w) = |\operatorname{supp} w|$ and $\mathbf{1} = \mathbf{1}_{[n]}$ is the all-ones word. For a set $T \subseteq [n]$ we write $\mathbf{1}_T$ for its indicator word. The (real) $0/1$ vector with the same entries is denoted by the same letter when no confusion can arise.

Definition 2.1 (Box)**.** A *box* is a pair $(M, [0,1]^n)$ where $M \in \mathbb{F}_2^{m \times n}$ is a binary matrix. Its *code* is $C = \ker M \subseteq \mathbb{F}_2^n$, its *nullity* is $k = \dim C = n - \operatorname{rank} M$, its *relation lattice* is

$$\Lambda_M = \{\alpha \in \mathbb{Z}^n : M\alpha \equiv 0 \pmod 2\},$$

and its *fibre* is $F = \Lambda_M \cap \{0,1\}^n \setminus \{0\}$. The columns $c_1, \dots, c_n \in \mathbb{F}_2^m$ of M are the *parity columns*.

Lemma 2.2. *The reduction map $\alpha \mapsto \alpha \bmod 2$ induces an isomorphism $\Lambda_M / 2\mathbb{Z}^n \cong C$. In particular Λ_M and the fibre depend only on the code C, and $F = C \setminus \{0\}$ under the identification of $0/1$ vectors with words. Every binary code $C \leq \mathbb{F}_2^n$ is the code of some box.*

Proof. $M\alpha \equiv 0 \pmod 2$ if and only if the reduction $\bar{\alpha}$ lies in $\ker M = C$; so Λ_M is the preimage of C under reduction, which contains $2\mathbb{Z}^n$. A $0/1$ vector is its own reduction. For the last assertion take M to be any matrix whose rows span the dual code $C^\perp$. □

Since only C matters, we shall state results for codes and speak of "the box C". The parity matrix is retained as the source of the name "parity columns"; a zero parity column $c_i = 0$ means $e_i \in C$.

Remark 2.3 (Four numbers). Four integers attached to a box must be kept apart: the *length* n, which is the ambient dimension of $\Lambda_M \subset \mathbb{Z}^n$ and of all cells; the *nullity* $k = \dim C$; the *reduced length* p (the number of coordinates after identifying coordinates that agree on every codeword, Section 6.2); and the *witness rank* of a collision (Definition 4.7). The minimum witness rank $r_{\min}$ is a minimum over collisions of the last quantity and is unrelated in size to the first three: Theorem 17.1 says $r_{\min} = 3$ for all n, k, p.

2.2 Occurrence coordinates

In the arithmetic setting a coordinate $i \in [n]$ corresponds to a prime (or prime-power) active in an occurrence, a lattice vector $\alpha \in \Lambda_M$ records the exponents, and the parity column c_i records how the ith coordinate enters the congruence defining the lattice. The box $[0,1]^n$ is the smallest box in which every coordinate can move. In this paper we only use the following facts about the model, all of which are definitional: (a) positions of the fibre are the nonzero words of C; (b) admissible moves between fibre points are Graver vectors of Λ_M; (c) a cell is a parallelepiped of independent admissible moves with all corners in the fibre. Two features of

the model are consequences of the choices made in GEO-I v006: the origin is excluded from the fibre ("deleted origin": an occurrence must have a nonzero exponent vector), and cells whose corner set would contain the origin are excluded.

Remark 2.4 (What is not claimed). Lemma 2.2 shows that the combinatorial model exists for every code. Whether the binary lattice Λ_M of a given code is the relation lattice of an actual rational occurrence fibre depends on which finite abelian groups and prime configurations can be realized arithmetically, a question that this paper does not address. Statements about binary boxes are statements about the frozen model.

3 Circuits, conformal minimality, Graver generators

3.1 Circuits of a code

Definition 3.1 (Circuit)**.** A *circuit* of a code C is a minimal (under inclusion) nonempty support of a word of C. We write $\mathcal{C}(C)$ for the set of circuits. Equivalently, a circuit is a minimal dependent set of columns of the parity matrix M (the circuits of the binary matroid represented by M [10]).

Since a support determines the word, we speak of a circuit as both a subset of $[n]$ and a word of C. A circuit of size 1 is a coordinate i with $e_i \in C$, equivalently with $c_i = 0$.

Lemma 3.2 (Basic properties)**.** *Let C be a code and $0 \neq w \in C$.*

(a) $\operatorname{supp} w$ *contains a circuit c, and $w \oplus c$ has support* $\operatorname{supp} w \setminus c$.

(b) w is a disjoint union of circuits.

(c) If c is a circuit and $v \in C$ with $\operatorname{supp} v \subseteq c$ *then $v \in \{0, c\}$. Two distinct circuits are never nested.*

Proof. (a) Among the nonzero words of C with support contained in $\operatorname{supp} w$ choose one of minimal support; any nonzero word with smaller support would also be contained in $\operatorname{supp} w$, so this support is a circuit c. Since $c \subseteq \operatorname{supp} w$ the symmetric difference is the set difference. (b) Induct on the weight: $w \oplus c$ is a word of smaller weight, either 0 or a disjoint union of circuits by induction. (c) is the definition of minimality. □

Definition 3.3 (Decomposition, partition)**.** A *circuit family* is a set $S = \{S_1, \dots, S_r\}$ of pairwise disjoint circuits, $r \geq 1$; its *union* is $U(S) = S_1 \cup \cdots \cup S_r$, which is the support of the word $\mathbf{1}_{U(S)} = S_1 \oplus \cdots \oplus S_r \in C$. We call S a *decomposition* of the word $\mathbf{1}_{U(S)}$ and a *partition of the set* $U(S)$ into circuits. For a set $U \subseteq [n]$ write $\Pi_U(C)$ for the set of partitions of U into circuits. If $\mathbf{1} \in C$ we write $\Pi = \Pi_{[n]}(C)$ for the set of partitions of the all-ones word; $\beta = |\Pi|$.

By Lemma 3.2(b), a set U has a partition into circuits if and only if $\mathbf{1}_U \in C$.

Lemma 3.4 (Span of a circuit family)**.** *Let $S = \{S_1, \dots, S_r\}$ be a circuit family. The words $S_1, \dots, S_r$ are linearly independent,* $\operatorname{span} S = \{\bigoplus_{j \in J} S_j : J \subseteq [r]\}$ *has 2^r elements, every element of* $\operatorname{span} S$ *is the indicator of a union of members of S, and $\mathbf{1}_{U(S)} \in \operatorname{span} S$.*

Proof. Disjoint nonzero words are independent, and a sum of disjoint words is their union. □

Lemma 3.5. *If $S \neq T$ are two partitions of the same set U into circuits, then neither $S \subseteq T$ nor $T \subseteq S$, and* $\operatorname{span} S$ *and* $\operatorname{span} T$ *are proper subspaces of the subspace $C_U = \{w \in C : \operatorname{supp} w \subseteq U\}$. In particular $|S|, |T| \geq 2$.*

Proof. If $S \subseteq T$ then $U = \bigcup S \subseteq \bigcup T = U$ forces equality of the unions of disjoint sets, so $S = T$. Next suppose $\operatorname{span} S = C_U$. Each $T_l \in T$ lies in $C_U = \operatorname{span} S$, so T_l is the union of some members of S (Lemma 3.4); it contains a member $S_j \subseteq T_l$, and since both are circuits $T_l = S_j$ (Lemma 3.2(c)). Hence $T \subseteq S$, contradicting the first assertion. Finally if $|S| = 1$ then $U = S_1$ is a circuit, so every circuit inside U equals U and $T = S$. □

3.2 Conformal order and the Graver basis of Λ_M

For $\alpha, \beta \in \mathbb{Z}^n$ write $\alpha \sqsubseteq \beta$ (α is *conformally smaller* than β) if $\alpha_i\beta_i \geq 0$ and $|\alpha_i| \leq |\beta_i|$ for all i. The *Graver basis* of a lattice $\Lambda \leq \mathbb{Z}^n$ is the set of $\sqsubseteq$-minimal elements of $\Lambda \setminus \{0\}$ [2, 12]. It is finite and it contains a conformal decomposition of every lattice vector.

Proposition 3.6 (Graver basis of Λ_M)**.** *The Graver basis of Λ_M consists of*

(i) all vectors $\varepsilon \in \{0, \pm 1\}^n$ whose support is a circuit of C (so each circuit c gives $2^{|c|}$ Graver vectors, one for every choice of signs), and

(ii) the vectors $\pm 2e_i$ for those coordinates i for which $\{i\}$ is not a circuit.

The proof is short and is given in Appendix B together with a discussion of the case of singleton circuits.

Proposition 3.7 (Cells of the $0/1$ box)**.** *Let $g_1, \dots, g_r$ ($r \geq 1$) be linearly independent Graver vectors of Λ_M and $x \in F$ such that all 2^r points $x + \sum_{j \in J} g_j$ ($J \subseteq [r]$) lie in F. Then there are pairwise disjoint circuits $S_1, \dots, S_r$ with $\operatorname{supp} g_j = S_j$, and*

$$g_{j,i} = 1 - 2x_i \quad (i \in S_j).$$

Moreover $x \in C \setminus \operatorname{span}\{S_1, \dots, S_r\}$. Conversely, for every circuit family $S = \{S_1, \dots, S_r\}$ and every $x \in C \setminus \operatorname{span} S$ the vectors $g_j = (1 - 2x_i)_{i \in S_j}$ (zero elsewhere) are independent Graver vectors with all corners $x + \sum_J g_j$ in F, and the corners are exactly the words of the coset $x + \operatorname{span} S$.

Proof. A vector $\pm 2e_i$ added to a $0/1$ vector leaves $\{0,1\}^n$, so every g_j is of type (i) in Proposition 3.6 with support a circuit S_j. Now $x + g_j \in \{0,1\}^n$ forces $g_{j,i} = 1 - 2x_i$ for all $i \in S_j$: if $x_i = 0$ then $g_{j,i} = +1$, if $x_i = 1$ then $g_{j,i} = -1$. If $S_j \cap S_l \ni i$ for $j \neq l$, the corner $x + g_j + g_l$ has ith entry $x_i + 2(1 - 2x_i) = 2 - 3x_i \notin \{0,1\}$, a contradiction; hence the supports are pairwise disjoint. The corner $x + \sum_{j \in J} g_j$ is the $0/1$ vector obtained from x by flipping the coordinates of $\bigcup_{j \in J} S_j$, i.e. it is $x \oplus \bigoplus_{j \in J} S_j$. All corners lie in $F \subseteq C \setminus \{0\}$; as $x \in C$ and $S_j \in C$, this is automatic for membership in C, and the corners are nonzero if and only if $x \notin \operatorname{span} S$. The converse is the same computation read backwards. □

Proposition 3.7 says that, in the $0/1$ box, the parallelepipeds spanned by Graver vectors with corners in the fibre are determined by the data (S, x) and are *exactly* the cells of the next section.

4 Binary cells, realizability, and non-embedding

4.1 Cells

Definition 4.1 (Cell)**.** For a circuit family $S = \{S_1, \dots, S_r\}$ and $x \in C \setminus \operatorname{span} S$ the *cell* $Z(S, x) \subseteq \mathbb{R}^n$ is

$$Z(S, x) = \Big\{ x + \sum_{j=1}^{r} t_j g_j : \ t \in [0,1]^r \Big\}, \qquad g_j = (1 - 2x_i)_{i \in S_j}.$$

Its *dimension* is r, its *union* is $U(S)$, its *vertices* are the 2^r words of the coset $x + \operatorname{span} S$ (all nonzero), and its *generator matrix* is the $r \times n$ integer matrix with rows g_j. A 0-*cell* is a single point of F.

Lemma 4.2 (Cells as pairs)**.** *$Z(S,x) = Z(T,y)$ if and only if $S = T$ and $x + y \in \operatorname{span} S$. Thus cells of positive dimension correspond bijectively to pairs $(S,\ x + \operatorname{span} S)$ with S a circuit family and $x + \operatorname{span} S$ a coset different from* span S.

Proof. The vertex set of $Z(S,x)$ is the coset $x + \operatorname{span} S$. A cell is a product (in the block coordinates S_j) of segments $[x|_{S_j},\ \bar{x}|_{S_j}]$ with $\bar{}$ denoting complement within the block; so the coordinates i at which the points of the cell are non-constant are $U(S)$, and the block structure (which coordinates move together with which) recovers S. The vertex set then gives the coset. The converse is clear: replacing x by $x \oplus s$, $s \in \operatorname{span} S$ replaces some g_j by $-g_j$ and shifts the base point by the corresponding sum. □

The cells of dimension $r \geq 1$ are called *admissible* (or *realized*) because all corners lie in the fibre F. A circuit family S admits a cell if and only if $\operatorname{span} S \neq C$; by Lemma 3.5 this holds whenever S is one of two distinct partitions of a set.

Lemma 4.3 (Relative interior)**.** *For $Z = Z(S,x)$ the relative interior is*

$$\operatorname{relint} Z = \Big\{ p \in \mathbb{R}^n :\ p_i = x_i \ (i \notin U(S)),\ \ p_i = x_i + t_j(1 - 2x_i) \text{ for } i \in S_j,\ t_j \in (0,1) \Big\}.$$

In particular every coordinate p_i with $i \in U(S)$ lies in the open interval $(0,1)$ and every coordinate $i \notin U(S)$ equals $x_i \in \{0,1\}$. The centre *of Z is the point $\gamma(U,z)$ with $\gamma_i = \frac{1}{2}$ for $i \in U$ and $\gamma_i = z_i$ for $i \notin U$, where $U = U(S)$ and $z = x|_{[n]\setminus U}$.*

Proof. The generators have disjoint supports, so $t \mapsto x + \sum t_j g_j$ is an affine isomorphism of $\mathbb{R}^r$ onto the affine hull of Z, mapping $(0,1)^r$ onto $\operatorname{relint} Z$. For $i \in S_j$ the coordinate is $x_i + t_j(1 - 2x_i)$, which equals t_j if $x_i = 0$ and $1 - t_j$ if $x_i = 1$; both lie in $(0,1)$ exactly when $t_j \in (0,1)$. At $t = (\frac{1}{2}, \ldots, \frac{1}{2})$ all coordinates in U equal $\frac{1}{2}$. □

Theorem 4.4 (Centre criterion)**.** `B2`. *Let $Z = Z(S,x)$ and $Z' = Z(T,y)$ be cells of positive dimension. Then* $\operatorname{relint} Z \cap \operatorname{relint} Z' \neq \emptyset$ *if and only if $U(S) = U(T)$ and $x_i = y_i$ for every $i \notin U(S)$. In that case the centre of Z equals the centre of Z' and lies in both relative interiors.*

Proof. If p lies in both relative interiors then, by Lemma 4.3, $\{i : 0 < p_i < 1\} = U(S) = U(T)$, and for i outside this set $p_i = x_i = y_i$. Conversely if $U(S) = U(T) = U$ and $x = y$ off U, then the point γ with $\gamma_i = \frac{1}{2}$ on U and $\gamma_i = x_i$ off U lies in both relative interiors (take all $t_j = \frac{1}{2}$). □

Source status. `B1` and `B2`: GEO-I v006, `OBSTRUCTION_RANK.md` §1 (PROVED); Proposition 3.7 restates `B1` for the $0/1$ box. Both are re-proved above from the definitions.

4.2 Collisions and embeddedness

Definition 4.5 (Collision; embedded)**.** Two *distinct* cells $Z \neq Z'$ of positive dimension *collide* if their relative interiors intersect; equivalently (Theorem 4.4) if $U(S) = U(T)$ and $x = y$ off U. A collision is a pair $\{Z, Z'\}$ of colliding cells. The box is *non-embedded* if it has a collision and *embedded* otherwise.

The word "embedded" is justified by the following description in terms of polyhedral complexes.

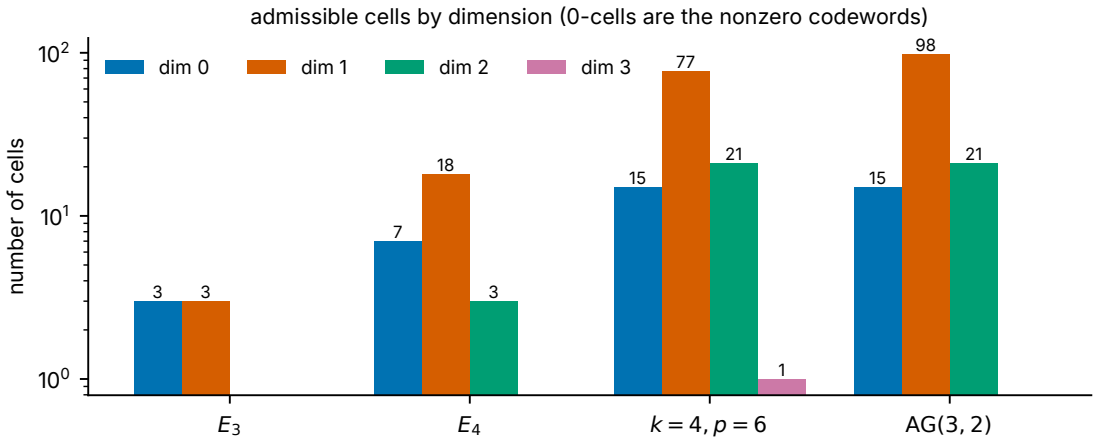


Figure 1: Numbers of cells of each dimension (dimension 0 counts the nonzero codewords) for E_3, E_4, a minimal class with $k = 4$, $p = 6$, and AG$(3, 2)$. A cell is determined by a family S of disjoint circuits and a coset of span S (Lemma 4.2). *Description:* exact combinatorial data computed by `scripts/figs.py`; no higher-dimensional geometry is implied by the drawing.

Proposition 4.6 (Embedded $\Leftrightarrow$ face-to-face)**.** *The faces of a cell $Z(S, x)$ are the cells $Z(S', x')$ with $S' \subseteq S$ and x' a vertex of Z, together with the vertices. The box is embedded if and only if any two cells (of any dimension) intersect in a common face, possibly empty; i.e. if and only if the cells form a polyhedral complex.*

Proof. The cell is a product of segments, one per block, so its faces are the sub-products: choose a subfamily $S' \subseteq S$ to vary and fix every other block at one of its two endpoints; the result is $Z(S', x')$ with x' a vertex (then $x' \notin \operatorname{span} S'$, because $x' \oplus s'$ is a vertex of Z, hence nonzero, for every $s' \in \operatorname{span} S' \subseteq \operatorname{span} S$). If a collision $\{Z, Z'\}$ exists, the common centre lies in $Z \cap Z'$; if $Z \cap Z'$ were a face G of both, then $G \ni \gamma$ with $\gamma \in \operatorname{relint} Z$ forces $G = Z$ (a face containing a point of the relative interior is the whole cell), and likewise $G = Z'$, so $Z = Z'$, a contradiction. Conversely suppose there is no collision, and let Z, Z' be cells with $p \in Z \cap Z'$. The unique face G of Z with $p \in \operatorname{relint} G$ and the unique face G' of Z' with $p \in \operatorname{relint} G'$ are cells or vertices; if G is a vertex then $p \in F$ is a $0/1$ point, and then G' is also a vertex because the relative interior of a positive-dimensional cell contains no $0/1$ points (Lemma 4.3); in the positive-dimensional case $\operatorname{relint} G \cap \operatorname{relint} G' \neq \emptyset$ and no collision forces $G = G'$. So every point of $Z \cap Z'$ lies in the relative interior of a common face. Taking p to be the average of points chosen in the relative interiors of the common faces of maximal dimension, the minimal face of Z containing p is a common face containing all of them; hence $Z \cap Z'$ is a single common face. □

4.3 Witnesses and ranks

Definition 4.7 (Witness; witness rank; $r_{\min}$)**.** Let $\{Z, Z'\}$ be a collision with generator matrices $A\,(r \times n)$ and $B\,(t \times n)$ and base points x, y; put $\delta = x - y$. The triple (A, B, δ) is a *witness*. Its *witness rank* is

$$d_{\text{span}} = \operatorname{rank} \begin{bmatrix} A \\ B \\ \delta \end{bmatrix}.$$

The *minimum witness rank* $r_{\min}(C)$ of a non-embedded box is the minimum of the witness rank over all collisions.

Lemma 4.8 (Witness rank is the dimension of the affine span)**.** *For a collision* $\{Z, Z'\}$ *the vector* δ *lies in the row space of* $\begin{bmatrix} A \\ B \end{bmatrix}$*. Consequently the witness rank equals* $\operatorname{rank}\begin{bmatrix} A \\ B \end{bmatrix}$ *and equals the dimension of the affine hull of* $Z \cup Z'$*. It is at least* $\max(r, t)$ *and at most* $r + t$*.*

Proof. Let γ be the common centre. Then $\gamma = x + \frac{1}{2}\sum g_j = y + \frac{1}{2}\sum h_l$ (rows g_j of A, h_l of B), so $\delta = x - y = \frac{1}{2}\sum h_l - \frac{1}{2}\sum g_j$. The affine hulls of Z and Z' are $\gamma + \operatorname{row}(A)$ and $\gamma + \operatorname{row}(B)$, so the hull of the union is $\gamma + \operatorname{row}(A) + \operatorname{row}(B)$. The rows of A are independent (disjoint supports), similarly for B. □

Remark 4.9 (Individual versus minimum). A collision may have witness rank larger than 3; the quantity $r_{\min}$ takes the minimum. For instance the even-weight code E_6 (Section 18) has 1710 collisions: 720 between cells of dimensions $(2, 2)$ and 990 between cells of dimensions $(3, 3)$, with witness ranks $3, 4, 5, 6$ occurring $585, 360, 585, 180$ times (exact enumeration with the engine of Appendix L), while the minimum is still 3. Only the minimum is bounded.

Example 4.10 (An embedded box, E_3)**.** Let $M = (1\,1\,1)$, $C = E_3 = \{000, 110, 101, 011\}$. The circuits are the three weight-two words, pairwise intersecting. A circuit family contains only one circuit, so all cells are 1-dimensional: the cell of the circuit c and base point $x \in C \setminus \{0, c\}$ joins the two remaining words (for $c = 110$, $x = 101$, the edge 101–011). The fibre is a triangle with three edges, the cells form a complex, and the box is embedded; Theorem 5.2 below shows this without enumeration, since no set has two partitions.

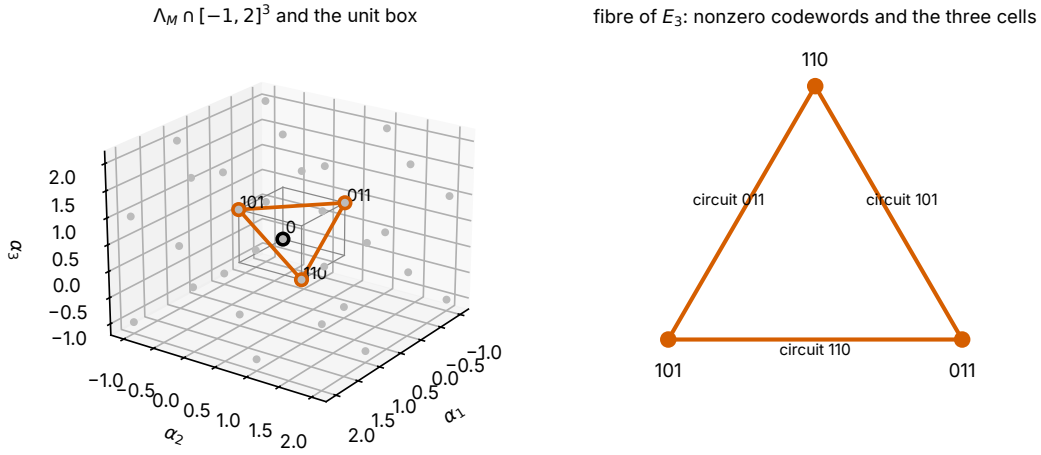


Figure 2: The binary relation lattice and its 0/1 slice for $n = 3$, $M = (1\,1\,1)$. Left: lattice points of Λ_M in $\{-1, 0, 1, 2\}^3$ (grey) and the unit box; the three nonzero fibre points $110, 101, 011$ are the vertices of the triangle (the edges are the three admissible cells). Right: the same data as codewords of E_3. *Description:* exact integer coordinates generated by `scripts/figs.py`; a rendering of $n = 3$ only, not of higher-dimensional geometry.

Part II
Circuit structure of non-embedded boxes

5 The exact circuit-partition criterion

Lemma 5.1. *A vector space* V *over* $\mathbb{F}_2$ *is not the union of two proper subspaces.*

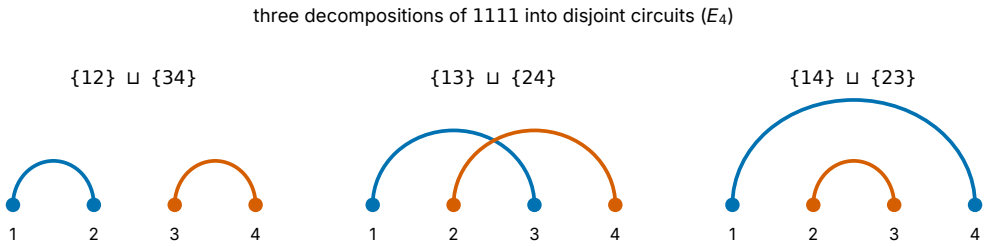


Figure 3: Circuits and decompositions in E_4. The word 1111 has three distinct decompositions into pairs of disjoint circuits. *Description:* generated by `scripts/figs.py` from the exact circuit list.

Proof. If $\dim V = d$ and $A, B \subsetneq V$ are subspaces, then $|A \cup B| \le |A| + |B| - 1 \le 2 \cdot 2^{d-1} - 1 < 2^d$ because $0 \in A \cap B$. □

(The statement is false for *three* subspaces over $\mathbb{F}_2$, as the three lines of a plane show. This is why the criterion below is formulated with *two* partitions.)

Theorem 5.2 (Circuit-partition criterion; `G10-E1`)**.** *For a code $C \le \mathbb{F}_2^n$ the following are equivalent.*

(i) The box C is non-embedded.

(ii) There is a set $U \subseteq [n]$ with $|\Pi_U(C)| \ge 2$, i.e. U has two distinct partitions into circuits.

(iii) Some word of C has two distinct decompositions into disjoint circuits.

Moreover, if $S \ne T$ are partitions of U into circuits and $x \in C \setminus (\operatorname{span} S \cup \operatorname{span} T)$ (such x exists), then $Z(S, x)$ and $Z(T, x)$ collide.

Proof. (ii)⇔(iii) is the identification of a word $\mathbf{1}_U$ with its support.

(ii)⇒(i). Let $S \ne T$ be partitions of U. By Lemma 3.5, $\operatorname{span} S$ and $\operatorname{span} T$ are proper subspaces of $C_U \le C$, hence of C. By Lemma 5.1 there is $x \in C \setminus (\operatorname{span} S \cup \operatorname{span} T)$. The cells $Z(S, x)$ and $Z(T, x)$ exist, have the same union U, agree (trivially) off U because they have the same base point, and are different because $S \ne T$ (Lemma 4.2). By Theorem 4.4 they collide.

(i)⇒(ii). Let $\{Z(S, x), Z(T, y)\}$ be a collision, so $U(S) = U(T) =: U$ and $x = y$ off U. If $S \ne T$ these are two distinct partitions of U. Suppose $S = T$. By Lemma 4.2 the cells are distinct, so $c := x \oplus y \notin \operatorname{span} S$; and $c \in C$ with $\operatorname{supp} c \subseteq U$ since $x = y$ off U. In particular $c \ne 0$. By Lemma 3.2(b) write $c = c_1 \sqcup \cdots \sqcup c_q$ as a disjoint union of circuits (all inside U). The word $\mathbf{1}_U \oplus c$ lies in C (because $\mathbf{1}_U = \bigoplus S_j \in C$) and has support $U \setminus \operatorname{supp} c$; if this is nonempty write it as a disjoint union of circuits $d_1 \sqcup \cdots \sqcup d_s$. Then $S' = \{c_1, \dots, c_q, d_1, \dots, d_s\}$ is a partition of U into circuits. If $S' = S$, every block of S would lie either inside $\operatorname{supp} c$ or inside $U \setminus \operatorname{supp} c$, so c would be a union of blocks of S, i.e. $c \in \operatorname{span} S$, which is false. So $S' \ne S$ and U has two distinct partitions. □

Source status. `G10-E1`, GEO-I v010, `MATHEMATICS.md` §1.1 (PROVED). The proof above uses only Lemmas 3.5 and 5.1 and Theorem 4.4. It was re-derived independently of the release text, and the equivalence (i)⇔(ii) was cross-checked by exhaustive computation on 4213 small codes against a direct enumeration of cells and relative-interior intersections (Appendix L); no discrepancy.

Remark 5.3 (Geometric content). Theorem 5.2 turns a metric–geometric property (overlap of relative interiors of parallelepipeds in $\mathbb{R}^n$) into a purely *linear-algebraic* one (non-uniqueness of a decomposition into minimal supports). Two features are worth recording. First, the collision produced in (ii)⇒(i) uses a *common* base point x for both cells; collisions of this kind are the ones to which the rank-three construction of Section 9 applies. Second, in (i)⇒(ii) a collision

with $S = T$ and distinct cosets forces a *third* partition: this is the only place where one needs that every nonzero word is a union of circuits.

Proposition 5.4 (Cells through a centre)**.** *Let $U \subseteq [n]$ with $\mathbf{1}_U \in C$, let $C_U = \{w \in C : \operatorname{supp} w \subseteq U\}$ and let $z \in \mathbb{F}_2^{[n]\setminus U}$ be such that $X_z = \{x \in C : x|_{[n]\setminus U} = z\}$ is nonempty. The number of cells of positive dimension whose union is U and whose base point agrees with z off U (that is, the cells whose relative interior contains the centre $\gamma(U, z)$) is*

$$N(U, z) = \sum_{S \in \Pi_U(C)} \Big(\frac{|C_U|}{2^{|S|}} - [z = 0] \Big).$$

In particular, if $\mathbf{1} \in C$, then the number of cells whose centre is $(\frac{1}{2}, \dots, \frac{1}{2})$ is $N = \sum_{P \in \Pi} (2^{k - m_P} - 1)$, with $m_P = |P|$ and $k = \dim C$, and the number of collisions with this centre is $\binom{N}{2}$.

Proof. X_z is a coset of C_U (the difference of two elements has support in U), so $|X_z| = |C_U|$. By Lemma 4.2 the cells over (S, z) correspond to cosets of $\operatorname{span} S \leq C_U$ in X_z, excluding the coset $\operatorname{span} S$ itself, which occurs inside X_z exactly when $z = 0$. The coset count is $|C_U|/2^{|S|}$ (Lemma 3.4). For $U = [n]$ the set X_z is C itself. □

Corollary 5.5 (Collisions of a minimal box)**.** *In a minimal non-embedded box every cell with $U = [n]$ has centre $\frac{1}{2}\mathbf{1}$, every collision has this centre (Theorem 7.1), and any two distinct cells with union $[n]$ collide. Hence the number of collisions is exactly $\binom{N}{2}$ with $N = \sum_{P \in \Pi} (2^{k - m_P} - 1)$.*

Proof. A collision has a common union U with two partitions, so $U = [n]$ by Theorem 7.1; for $U = [n]$ the condition "equal off U" is empty, so any two distinct cells with union $[n]$ collide (Theorem 4.4). □

Source status. New here: this closed formula was not found in the release files read. It was verified by direct enumeration of cells for the $k = 3, 4, 5$ classes (Appendix L); e.g. E_4 has $N = 3$ (3 collisions), the $k = 4$, $p = 6$ box has $N = 13$ (78 collisions) and the $k = 5$, $p = 8$ box has $N = 57$ (1596 collisions), in agreement with the direct enumeration of all collisions (Section 18).

6 Minimal non-embedded boxes and shortening

6.1 Shortening

Definition 6.1 (Shortening)**.** For $i \in [n]$ the *shortening* of C at i is the code $C_{(i)} = \{w|_{[n]\setminus\{i\}} : w \in C,\ w_i = 0\} \leq \mathbb{F}_2^{n-1}$. It is the kernel of the parity matrix with the ith column deleted. For $I \subseteq [n]$ write $C_{(I)}$ for the iterated shortening. The box $C_{(i)}$ is called a *sub-box* of C.

Deleting a column of M corresponds geometrically to fixing the coordinate i of every lattice vector at 0: the lattice $\Lambda_{M'}$ of the deleted matrix is the intersection of Λ_M with $\{\alpha_i = 0\}$, and the box $[0, 1]^{n-1}$ is the corresponding face of $[0, 1]^n$.

Lemma 6.2 (Circuits and cells of a shortening)**.** *(a) The circuits of $C_{(i)}$ are exactly the circuits of C not containing i. (b) A circuit family of $C_{(i)}$ is a circuit family of C avoiding i; the cells of $C_{(i)}$ are the cells $Z(S, x)$ of C with $i \notin U(S)$ and $x_i = 0$ (with coordinate i deleted). (c) Two cells of $C_{(i)}$ collide if and only if the corresponding cells of C collide, and the generator matrices of the pair differ only by an added zero column; so the witness ranks agree.*

Proof. (a) A word of $C_{(i)}$ is a word of C vanishing at i. If c is a circuit of C with $i \notin c$ then every word of C with support inside c vanishes at i, so minimality in C gives minimality in $C_{(i)}$. Conversely a circuit of $C_{(i)}$ is the support of a word of C vanishing at i, and every word of C with smaller support also vanishes at i, so it is a circuit of C avoiding i. (b) follows from (a) and the definition of cells, since $x \in C_{(i)}$ means $x \in C$, $x_i = 0$. (c) The condition "same union and equal off the union" reads the same in $\mathbb{F}_2^{n-1}$ and in $\mathbb{F}_2^n$ when $x_i = y_i = 0$ and i lies outside the common union. □

Definition 6.3 (Minimal box)**.** A box C is *minimal* (more fully: *coordinatewise-minimal non-embedded*) if C is non-embedded and $C_{(i)}$ is embedded for every $i \in [n]$.

Proposition 6.4 (Descent to a minimal sub-box)**.** *Every non-embedded box C has a minimal sub-box $C_{(I)}$, $I \subseteq [n]$, and $r_{\min}(C) \leq r_{\min}(C_{(I)})$.*

Proof. If C is non-embedded and not minimal, some $C_{(i)}$ is non-embedded and has length $n-1$. Repeat; the process stops because the length decreases at every step. By Lemma 6.2(c) every collision of $C_{(I)}$ lifts to a collision of C with the same witness rank, so $r_{\min}(C) \leq r_{\min}(C_{(I)})$. □

Source status. `G9-T5` (reduction to minimal boxes), GEO-I v009 `MATHEMATICS.md` §2.4 (PROVED+CHECKED); the argument above is the general reduction without reference to nullity.

Corollary 6.5 (Nullity at least three)**.** *If C is non-embedded then $k = \dim C \geq 3$.*

Proof. By Theorem 5.2 there are partitions $S \neq T$ of some U. By Lemma 3.5, $2 \leq |S| = \dim \operatorname{span} S < \dim C_U \leq k$, so $k \geq 3$. □

Source status. `B-NULL′`, GEO-I v008 `MINIMAL_RANK.md` (PROVED): any collision forces nullity ≥ 3.

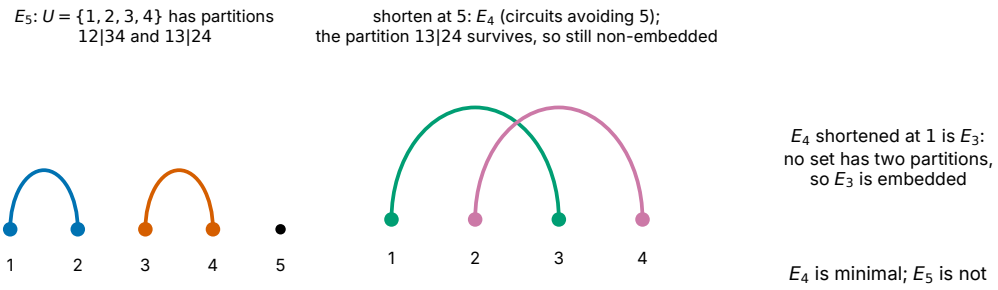


Figure 4: Shortening a coordinate: deleting a parity column keeps exactly the circuits avoiding that coordinate. *Description:* exact combinatorial data computed by `scripts/figs.py`; no higher-dimensional geometry is implied by the drawing.

6.2 Series classes and reduced boxes

Definition 6.6 (Series classes)**.** Coordinates i, j are *series-equivalent* if $w_i = w_j$ for all $w \in C$. A coordinate is *dead* if $w_i = 0$ for all $w \in C$. The *reduced length* $p(C)$ is the number of series classes of non-dead coordinates; the *multiplicity* of a class is its size. The *reduced code* $\bar{C} \leq \mathbb{F}_2^p$ is obtained by keeping one coordinate from every class of non-dead coordinates (and deleting dead ones); C is a *blow-up* of $\bar{C}$ (plus dead coordinates).

(In the generator-matrix picture of Section 8, series-equivalent coordinates are the equal columns, and dead coordinates are the zero columns of a generator matrix; "series" and "parallel" are matroid names for the same relation, depending on whether one reads the code or its dual [10]. We use "series".)

Proposition 6.7 (Blow-up invariance; `B-SERIES`)**.** *Let C have no dead coordinates. The words of C are constant on series classes, and replacing every class by a single coordinate gives an isomorphism $C \cong \bar{C}$ of codes. Under this isomorphism:*

(a) the circuits of C are the preimages of the circuits of $\bar{C}$ (so every circuit is a union of classes);

(b) cells correspond to cells, collisions to collisions; the generator matrices differ by duplicating columns, so witness ranks coincide and $r_{\min}(C) = r_{\min}(\bar{C})$;

(c) C is non-embedded iff $\bar{C}$ is; and C is minimal iff $\bar{C}$ is minimal.

If C has dead coordinates, the dead coordinates are in no circuit and in no cell; C is non-embedded iff the code without the dead coordinates is, and then C is not minimal.

Proof. Constancy on classes holds by definition and gives the isomorphism. Supports correspond to unions of classes and inclusion is preserved, so (a) follows; (b) follows because union, equality off the union, and the cell data (S, x) all transfer, and duplicating columns of a matrix does not change its row rank. For (c): non-embeddedness transfers by (b). For minimality note that the circuits of $C_{(i)}$ are the circuits of C avoiding i (Lemma 6.2); these are the preimages of the circuits of $\bar{C}$ avoiding the class of i, i.e. the circuits of $\bar{C}_{(\bar{\imath})}$; by Theorem 5.2 embeddedness of a shortening depends only on its circuits and the sets they form, so $C_{(i)}$ is embedded iff $\bar{C}_{(\bar{\imath})}$ is. The statement on dead coordinates holds since a dead coordinate lies in no support, and $C_{(i)}$ for a dead i is the code with i deleted. □

Source status. `B-SERIES`, GEO-I v008 `MINIMAL_RANK.md` (PROVED+CHECKED). Reproved above via Theorem 5.2.

Remark 6.8. A blow-up of a minimal box is minimal, although shortening a coordinate of a class of size ≥ 2 turns the other coordinates of the class into dead ones (zero columns of the generator matrix). This is consistent with Proposition 6.7 because dead coordinates are irrelevant for embeddedness. Hence the classification of minimal boxes reduces to the classification of reduced minimal boxes plus multiplicities, and all rank statements for reduced boxes carry over to all lengths n (Section 19).

7 The full-support theorem and zero-column exclusion

Theorem 7.1 (Full support; `G10-F1`)**.** *Let C be a minimal non-embedded box. If $U \subseteq [n]$ has two distinct partitions into circuits then $U = [n]$. Consequently $\mathbf{1} \in C$, the set $[n]$ has at least two partitions into circuits, and every coordinate lies in a circuit.*

Proof. Let $S \neq T$ be partitions of U and suppose $i \notin U$. All circuits of S and T avoid i, so by Lemma 6.2(a) they are circuits of $C_{(i)}$ and $S \neq T$ are two partitions of U in $C_{(i)}$. By Theorem 5.2, $C_{(i)}$ is non-embedded, contradicting minimality. By Theorem 5.2 some U has two partitions; hence $U = [n]$ is the only such set. The word $\mathbf{1} = \mathbf{1}_{[n]}$ is then a sum of circuits, hence lies in C. □

Source status. `G10-F1`, GEO-I v010 `MATHEMATICS.md` §1.2 (PROVED); closes the earlier conjecture `G8-C3` (status CORRECTED: formerly open, now proved). This is an immediate consequence of the shortening lemma and Theorem 5.2.

Proposition 7.2 (No singleton circuits; `G9-Z`, `G11-L5c`)**.** *If $e_i \in C$ for some i (that is, $\{i\}$ is a circuit; equivalently the ith parity column vanishes) and C is non-embedded, then $C_{(i)}$ is non-embedded. Consequently a minimal non-embedded box has no circuit of size* 1*: every circuit has at least two coordinates, and there is no zero parity column.*

Proof. Let $S \neq T$ be partitions of a set U (Theorem 5.2). If $i \notin U$ both are partitions in $C_{(i)}$ and we are done by Lemma 6.2. If $i \in U$, let $S_j \ni i$ be the block of S containing i. Since $\{i\}$ is a circuit and circuits are not nested (Lemma 3.2(c)), $S_j = \{i\}$; likewise $\{i\} \in T$. Then $S \setminus \{\{i\}\}$ and $T \setminus \{\{i\}\}$ are two different partitions of $U \setminus \{i\}$ (different because $S \neq T$; if one of them is empty then $S = \{\{i\}\}$, and T, being a partition of the same set containing $\{i\}$, would equal it). Both avoid i, so $C_{(i)}$ is non-embedded. □

Source status. `G9-Z`, GEO-I v009 `MATHEMATICS.md` §1 (PROVED+CHECKED); `G11-L5c`, GEO-I v011 `MATHEMATICS.md` §1.1 (PROVED). The two statements in the release files are proved by separate arguments; the single argument above gives both.

Corollary 7.3 (Properties of a minimal box)**.** *A minimal non-embedded box C of nullity k satisfies: (P1) $\mathbf{1} \in C$ and $|\Pi| \geq 2$; (P2) no coordinate is dead; every circuit has at least two coordinates; (P3) $k \geq 3$ and $\mathbf{1}$ is not a circuit; (P4) every partition of $\mathbf{1}$ has $m_P \geq 2$ blocks of size ≥ 2 each.*

Proof. (P1) Theorem 7.1. (P2) $\mathbf{1} \in C$ has full support, and Proposition 7.2. (P3) Corollary 6.5; if $\mathbf{1}$ were a circuit then every circuit equals $\mathbf{1}$ and Π has a single element. (P4) is (P2) together with (P3). □

8 Affine-geometric reduction

8.1 Generator matrices and point sets

Let $C \leq \mathbb{F}_2^n$ be a code of dimension k without dead coordinates. Choose a basis $b_1, \ldots, b_k$ of C and let G be the $k \times n$ matrix with rows b_s. Its columns $g_1, \ldots, g_n \in \mathbb{F}_2^k \setminus \{0\}$ satisfy

$$w_f := \big(f \cdot g_i\big)_{i \in [n]} \in C, \qquad C = \{w_f : \ f \in \mathbb{F}_2^k\}, \qquad f \mapsto w_f \text{ bijective.}$$

Thus a code of dimension k without dead coordinates is a *multiset* of nonzero points of F_2^k spanning $\mathbb{F}_2^k$ (the column multiset), up to the action of $\mathrm{GL}(k,2)$ (change of basis of C) and permutation of coordinates. Series classes are the sets of coordinates with equal columns; the reduced code corresponds to the set $X = \{g_i\}$ of *distinct* columns, $|X| = p$, spanning $\mathbb{F}_2^k$.

Lemma 8.1. *Two reduced codes $\bar{C}, \bar{C}'$ with point sets $X, X' \subseteq \mathbb{F}_2^k \setminus \{0\}$ (spanning) are equal up to a permutation of coordinates if and only if $X' = \sigma X$ for some $\sigma \in \mathrm{GL}(k,2)$.*

Proof. If $\pi \bar{C} = \bar{C}'$ for a coordinate permutation π, then $G'\Pi = AG$ for an invertible A (two generator matrices of the same code differ by a change of basis), so the column of G' at position $\pi(i)$ is Ag_i: $X' = AX$. Conversely if $X' = \sigma X$ then G', whose columns are σg_i in some order, generates $\{f \cdot \sigma g_i\} = \{(\sigma^\top f) \cdot g_i\}$, the same code up to the coordinate order. □

8.2 The affine hyperplane

Since $\mathbf{1} \in C$ for minimal boxes (Theorem 7.1) there is a unique $f_0 \in \mathbb{F}_2^k$ with $w_{f_0} = \mathbf{1}$, i.e. $f_0 \cdot g_i = 1$ for all i. The set $H = \{x \in \mathbb{F}_2^k : f_0 \cdot x = 1\}$ is an affine hyperplane, in bijection with $\mathbb{F}_2^{k-1}$ after choosing coordinates in which $f_0 = e_k^*$, and its affine structure is that of the affine space $\mathrm{AG}(k-1,2)$ (the points of H are the vectors with last coordinate 1). The subgroup of $\mathrm{GL}(k,2)$ preserving H (equivalently fixing f_0) acts on H as the full affine group $\mathrm{AGL}(k-1,2)$.

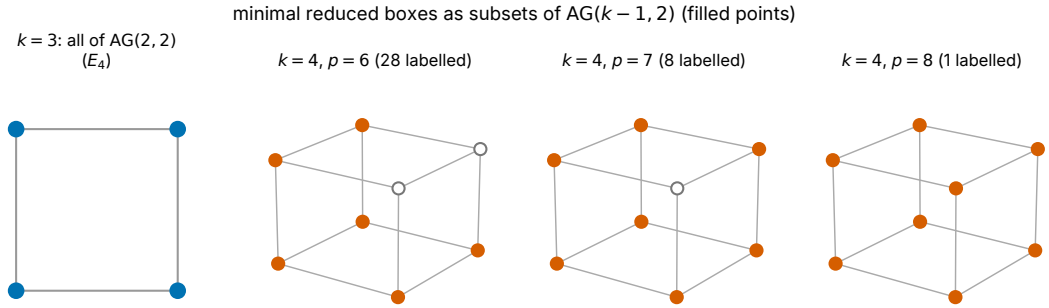


Figure 5: Affine reduction: the reduced coordinate set as a point set in an affine space. *Description:* exact combinatorial data computed by `scripts/figs.py`; no higher-dimensional geometry is implied by the drawing.

Theorem 8.2 (Affine reduction; `G10-A1`)**.** *Let C be a minimal non-embedded box of nullity k and $X \subseteq \mathbb{F}_2^k$ its reduced point set, $p = |X|$. Then:*

(a) $X \subseteq H \cong \mathrm{AG}(k-1,2)$ and X spans $\mathbb{F}_2^k$ (equivalently, X is not contained in any proper affine subspace of H);

(b) $p \geq k+1$; and $p = k+1$ holds if and only if $C = E_p$ (and then $p = 4$, $k = 3$);

(c) two minimal boxes have isomorphic reduced codes iff their reduced point sets lie in the same $\mathrm{AGL}(k-1,2)$-orbit of subsets of H.

Conversely, for every spanning subset $X \subseteq H$ the code $C(X) = \{(f \cdot x)_{x \in X}\}$ has dimension k and contains $\mathbf{1}$; whether it is non-embedded, and whether it is minimal, depends on X only through its $\mathrm{AGL}(k-1,2)$-orbit.

Proof. (a) $\mathbf{1} \in C$ gives $X \subseteq H$; X spans because the generator matrix has rank k; and "spans $\mathbb{F}_2^k$" means that X is not contained in a proper linear subspace; since $X \subset H$ and $0 \notin H$, this means X is not contained in a proper affine subspace of H.

(b) $p = k$ would make X a basis, so $C = \mathbb{F}_2^p$ with all singletons as circuits, contradicting Proposition 7.2. For $p = k+1$ the relations among the columns form a 1-dimensional space $\langle a \rangle = C^\perp$, hence $C = a^\perp$. If $a_i = 0$ then $e_i \in C$, a singleton circuit; so $a = \mathbf{1}$, $C = \mathbf{1}^\perp = E_p$. (The circuits of E_p are the weight-two words, and E_p is minimal only for $p = 4$, Proposition 8.3.)

(c) Lemma 8.1 gives equivalence under $\mathrm{GL}(k,2)$. If $\sigma X = X'$ and both sets lie in H, then σ maps X into H; because X spans, $f_0 \circ \sigma^{-1}$ takes the value 1 on X', which spans, and is a functional equal to 1 on a spanning subset of H, so $f_0 \circ \sigma^{-1} = f_0$ (a functional is determined by its values on a spanning set). So σ lies in the stabiliser of f_0, which acts on H as $\mathrm{AGL}(k-1,2)$.

The final statement is Lemma 8.1 together with the fact that isomorphic codes have the same circuits and hence the same embeddedness and minimality (Theorem 5.2). □

Source status. `G10-A1`, GEO-I v010 `MATHEMATICS.md` §1.3 (PROVED). The statements on $p \geq k+1$ and on E_p are made explicit here.

Proposition 8.3 (The even-weight codes)**.** *For $p \geq 3$ let $E_p = \{w \in \mathbb{F}_2^p : \mathrm{wt}(w) \text{ even}\}$. Its circuits are the $\binom{p}{2}$ words of weight two. E_3 is embedded, E_p is non-embedded for $p \geq 4$, E_4 is minimal, and E_p is not minimal for $p \geq 5$.*

Proof. Every even word of weight ≥ 4 contains a weight-two word, so the circuits are the pairs. A family of disjoint pairs is a partial matching; E_3 has no two disjoint pairs, so the

families are single pairs and every set has at most one partition: embedded. For $p \geq 4$ the set $\{1,2,3,4\}$ has the partitions $\{12,34\}$ and $\{13,24\}$: non-embedded. Shortening E_p at a coordinate gives E_{p-1} (the words vanishing at i are the even words on the other coordinates), so E_4 has the embedded shortening E_3, while for $p \geq 5$ the shortening E_{p-1} with $p-1 \geq 4$ is non-embedded. □

8.3 Consequences

Theorem 8.2 reduces the study of minimal boxes to the study of subsets of the finite affine spaces $\mathrm{AG}(k-1,2)$ up to affine equivalence. It also fixes the extreme case: the full affine space $X = \mathrm{AG}(k-1,2)$, which gives the first-order Reed–Muller code $\mathrm{RM}(1,k-1)$ of length 2^{k-1} (Proposition 15.2). All the arithmetic below is done with k (the dimension of the ambient space $\mathbb{F}_2^k$), p (the number of points) and the partition data of X.

9 Collision rank and local rank-three extraction

Lemma 9.1 (Lower bound; `B-LOW`)**.** *Every collision has witness rank at least* 3*. Consequently* $r_{\min}(C) \geq 3$ *for every non-embedded box.*

Proof. Let $\{Z(S,x), Z(T,y)\}$ be a collision, $r = |S|, t = |T|$, with generators g_j and h_l. By Lemma 4.8 the witness rank is at least $\max(r,t)$, so we may assume $r,t \leq 2$.

Case $r = 1$. Then $U = S_1$ is a circuit, so every circuit inside U equals U and $T = \{U\} = S$. As $x = y$ off U, the word $c = x \oplus y$ has support in U, so $c \in \{0, U\} = \operatorname{span} S$ (Lemma 3.2(c)), and $Z(S,x) = Z(S,y)$ by Lemma 4.2, a contradiction. The case $t = 1$ is symmetric. So $r = t = 2$.

Case $r = t = 2$. Suppose the witness rank equals 2. Then the rows of B lie in the row space of A: $h_l = \lambda_{l1} g_1 + \lambda_{l2} g_2$ with real λ. Since g_j has entries ± 1 exactly on S_j and h_l has entries in $\{0, \pm 1\}$, $|\lambda_{lj}| \in \{0,1\}$. If both were nonzero then $T_l = S_1 \cup S_2 = U$ and the other member of T could not be disjoint from it; so each T_l equals S_1 or S_2, and $T = S$. Then $h_l = \pm g_l$ means $1 - 2y_i = \varepsilon_l(1 - 2x_i)$ on S_l, i.e. $y|_{S_l} = x|_{S_l}$ or $y|_{S_l} = \bar{x}|_{S_l}$ for each l; hence $c = x \oplus y$ is constant on every block, so $c \in \operatorname{span} S$ (it vanishes off U because $x = y$ there), and the cells coincide: a contradiction. So the rank is at least 3. □

Source status. `B-LOW` (also `G8-T3a`), GEO-I v008 `MINIMAL_RANK.md` (PROVED).

Theorem 9.2 (Local rank-three extraction; `G10-L5`, `B-COMMON`)**.** *Let* $a \in C$ *be a word with two distinct decompositions into two disjoint circuits,* $a = b_1 \sqcup b_2 = c_1 \sqcup c_2$ *with* $\{b_1, b_2\} \neq \{c_1, c_2\}$*. Put* $S = \{b_1, b_2\}$, $T = \{c_1, c_2\}$ *and choose any* $x \in C \setminus (\operatorname{span} S \cup \operatorname{span} T)$*. Then* $Z(S,x)$ *and* $Z(T,x)$ *collide, and their four generators satisfy*

$$g_1 + g_2 = h_1 + h_2 = \big((1 - 2x_i)\mathbf{1}_{i \in \operatorname{supp} a}\big)_{i \in [n]},$$

so the witness rank is exactly 3*. In particular* $r_{\min}(C) = 3$ *for any code that contains such a word.*

Proof. Theorem 5.2 gives the collision. On $U = \operatorname{supp} a$ the generators of the two cells are the restrictions of the same vector $s = (1 - 2x_i)_i$ to the blocks: $g_j = s|_{b_j}$, $h_l = s|_{c_l}$. Hence $g_1 + g_2 = s|_U = h_1 + h_2$, a nontrivial linear relation among four vectors; their span has dimension at most 3. Lemma 9.1 gives at least 3. □

Source status. `G10-L5` (= `G10-T5`) and `B-COMMON`, GEO-I v010 `MATHEMATICS.md` §1.5 and v008 `MINIMAL_RANK.md` (PROVED).

Definition 9.3 (Condition W2)**.** A box satisfies *W2* if $\mathbf{1} \in C$ and the set $\Pi^{(2)}$ of partitions of $\mathbf{1}$ into exactly two circuits has at least two elements; $n_2 = |\Pi^{(2)}|$.

Theorem 9.4 (Conditional universality; `G10-C1′`)**.** *If every minimal non-embedded box satisfies W2, then* $r_{\min}(C) = 3$ *for* every *non-embedded box C.*

Proof. By Proposition 6.4 there is a minimal sub-box $C_{(I)}$ with $r_{\min}(C) \le r_{\min}(C_{(I)})$. By W2 the word $\mathbf{1} \in C_{(I)}$ has two distinct two-circuit decompositions, so Theorem 9.2 gives $r_{\min}(C_{(I)}) = 3$. Hence $r_{\min}(C) \le 3$, and Lemma 9.1 gives $r_{\min}(C) \ge 3$. □

Source status. `G10-C1′`, GEO-I v010 `MATHEMATICS.md` §1.5 (PROVED). The hypothesis W2 for *all* minimal boxes is the content of Theorem 16.3 (Part III), proved unconditionally in this paper.

Theorem 9.4 thus reduces the universal rank statement to a purely combinatorial one: *every minimal non-embedded box has two distinct partitions of the all-ones word into two circuits*. Part III proves this.

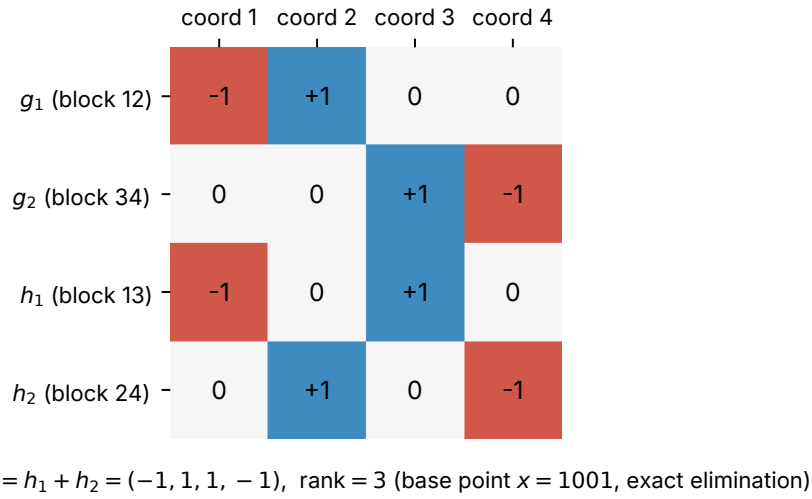


Figure 6: A rank-three collision in E_4 with common base point $x = 1001$. The four generators of the cells $Z(\{12, 34\}, x)$ and $Z(\{13, 24\}, x)$ as a 4×4 matrix, and the relation $g_1 + g_2 = h_1 + h_2 = (-1, 1, 1, -1)$. *Description:* exact integer data; rank computed by exact elimination, `scripts/figs.py`.

Part III
Rigidity, vector-space partitions and universality

Throughout Part III, C is a *minimal non-embedded box* of nullity k. We use the reduced code (Proposition 6.7): p denotes the number of coordinates of the reduced code, which we identify with C, so blocks of partitions of $\mathbf{1}$ have sizes measured in the reduced code and $\sum_{B \in P} |B| = p$. All statements in this part are statements about the reduced code; by Proposition 6.7 they transfer to blow-ups.

10 Circuit rigidity

Theorem 10.1 (Rigidity; `G10-R`)**.** *Let C be a minimal non-embedded box. Then:*

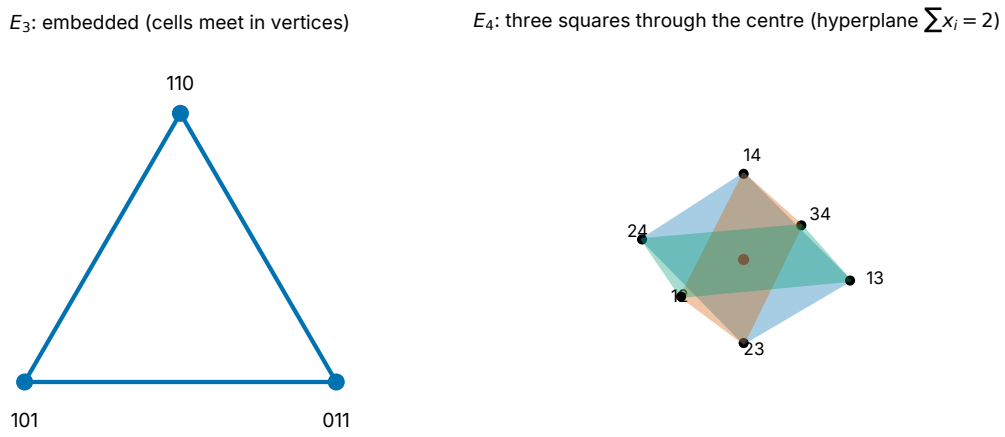


Figure 7: Embedded versus non-embedded. Left: E_3 (a triangle; cells meet in vertices). Right: the octahedron of the six weight-two words of E_4 in the hyperplane $\sum x_i = 2$ with the three equatorial squares, which share diagonals and all contain the centre. *Description:* an exact subcomplex of the 4-dimensional cell structure; the vertex 1111 and the cells through it are omitted. Not a picture of the full cell complex.

(a) every word $w \in C \setminus \{0, \mathbf{1}\}$ has a unique *decomposition into pairwise disjoint circuits;*

(b) the word $\mathbf{1}$ has $\beta = |\Pi| \geq 2$ decompositions;

(c) the number of circuit families (nonempty sets of pairwise disjoint circuits) of C is $2^k - 2 + \beta$, and a family whose union is not $[n]$ is determined by its union.

Proof. (a) If $w \neq 0, \mathbf{1}$ had two decompositions $S \neq T$, the set $U = \operatorname{supp} w \neq [n]$ would have two partitions, contradicting Theorem 7.1. (b) is Theorem 7.1. (c) Families with union $\neq [n]$ correspond bijectively to words $w \in C \setminus \{0, \mathbf{1}\}$ (each such word has exactly one decomposition, by (a), and each family determines its union word); there are $2^k - 2$ such words. The families with union $[n]$ are the β elements of Π. □

Source status. `G10-R`, GEO-I v010 `MATHEMATICS.md` §1.4 (PROVED): rigidity and the count of families. The proof above is a direct consequence of Theorem 7.1.

The converse also holds and shows that rigidity is not just a consequence but a characterisation.

Proposition 10.2 (Rigidity characterises minimality)**.** *A code C with $\mathbf{1} \in C$ is minimal non-embedded if and only if $\mathbf{1}$ has at least two decompositions into disjoint circuits and every $w \in C \setminus \{0, \mathbf{1}\}$ has exactly one.*

Proof. ($\Rightarrow$) Theorem 10.1. ($\Leftarrow$) C is non-embedded by Theorem 5.2. Let $i \in [n]$. A nonzero word of $C_{(i)}$ is a word $w \in C$ with $w_i = 0$, so $w \neq \mathbf{1}$ and w has a unique decomposition into circuits of C; these circuits lie in $\operatorname{supp} w$, so they avoid i and are circuits of $C_{(i)}$ (Lemma 6.2). Hence no word of $C_{(i)}$ has two decompositions in $C_{(i)}$, and $C_{(i)}$ is embedded by Theorem 5.2. □

Source status. The converse is noted here; it was not found stated in the release files read.

The bijection of words and circuit families in Theorem 10.1 makes the combinatorics of a minimal box unusually rigid: the family structure is that of a *linear space with exactly one failure of unique factorisation*, and the next section shows that this single failure is organised by a vector-space partition.

11 Vector-space partitions from partitions of the all-ones word

Recall that Π is the set of partitions P of $[n]$ (in the reduced code, of the p coordinates) into circuits; $m_P = |P| \geq 2$ is the number of blocks (Corollary 7.3), and each block has size ≥ 2. Put

$$V_P = \operatorname{span} P = \Big\{ \bigoplus_{B \in J} B : \ J \subseteq P \Big\} \leq C, \qquad \dim V_P = m_P, \qquad \{0, \mathbf{1}\} \subseteq V_P.$$

(Lemma 3.4.)

Lemma 11.1 (Existence)**.** *Every $w \in C \setminus \{0, \mathbf{1}\}$ lies in V_P for at least one $P \in \Pi$.*

Proof. Both w and $\mathbf{1} \oplus w$ are nonzero words of C (neither equals $\mathbf{1}$ or 0 because $w \neq 0, \mathbf{1}$). Decompose each into disjoint circuits (Lemma 3.2(b)); their supports are complementary, so the union of the two decompositions is a partition P of $[n]$ into circuits, and w is the sum of the blocks coming from the decomposition of w: $w \in V_P$. □

Lemma 11.2 (Uniqueness)**.** *If $w \in V_P \cap V_Q$ with $w \neq 0, \mathbf{1}$ then $P = Q$.*

Proof. Write $w = \bigsqcup_{B \in J} B$ with $J \subseteq P$ and $w = \bigsqcup_{B' \in J'} B'$ with $J' \subseteq Q$. Both are decompositions of w into disjoint circuits, so by rigidity (Theorem 10.1(a)) $J = J'$ as sets of circuits. Likewise $\mathbf{1} \oplus w$ is the union of $P \setminus J$ and of $Q \setminus J'$, so $P \setminus J = Q \setminus J'$. Hence $P = J \cup (P \setminus J) = J' \cup (Q \setminus J') = Q$. □

Theorem 11.3 (Vector-space partition; `G11-L5`)**.** *Let C be a minimal non-embedded box of nullity k. The subspaces V_P ($P \in \Pi$) satisfy $V_P \cap V_Q = \{0, \mathbf{1}\}$ for $P \neq Q$ and $\bigcup_P V_P = C$. Consequently the subspaces*

$$\bar{V}_P = V_P / \langle \mathbf{1} \rangle \ \leq \ C / \langle \mathbf{1} \rangle \cong \mathbb{F}_2^{k-1}$$

of dimension $m_P - 1 \geq 1$ form a vector-space partition *of $\mathbb{F}_2^{k-1}$: every nonzero vector lies in exactly one $\bar{V}_P$. A part has dimension 1 if and only if P has exactly two blocks.*

Proof. For $w \neq 0, \mathbf{1}$ Lemmas 11.1 and 11.2 give exactly one P with $w \in V_P$; also $0, \mathbf{1} \in V_P$ for all P, so $V_P \cap V_Q = \{0, \mathbf{1}\}$ and the union is C. Passing to $C / \langle \mathbf{1} \rangle$, the images of $V_P \setminus \{0, \mathbf{1}\}$ are pairwise disjoint and cover the nonzero classes; dimension is $m_P - 1$. □

Source status. `G11-L5`, GEO-I v011 `MATHEMATICS.md` §1.2–1.3 (PROVED). The terminology follows the standard notion of a vector-space partition [3]: a family of subspaces such that each nonzero vector lies in exactly one member (only the abstract page of the survey was read).

Remark 11.4 (Type). The multiset $\{m_P - 1 : P \in \Pi\}$ is the *type* of the partition in the usual sense. For example the minimal reduced box with $k = 4$, $p = 6$ has four parts of dimension 1 and one of dimension 2 (four two-block partitions of $\mathbf{1}$ and one three-block partition); for E_4 the type is $[1^3]$ in $\mathbb{F}_2^2$; for $X = \mathrm{AG}(k-1, 2)$ it is $[1^{2^{k-1}-1}]$, the trivial partition of $\mathbb{F}_2^{k-1}$ into its $2^{k-1} - 1$ points.

12 The counting identity

Theorem 12.1 (Counting identity; `G11-L5`)**.** *For a minimal non-embedded box of nullity k,*

$$\sum_{P \in \Pi} \big(2^{m_P} - 2\big) = 2^k - 2, \qquad \textit{equivalently} \qquad \sum_{P \in \Pi} \big(2^{m_P - 1} - 1\big) = 2^{k-1} - 1.$$

In terms of $n_m = \#\{P : m_P = m\}$: $\sum_{m \geq 2} n_m (2^m - 2) = 2^k - 2$.

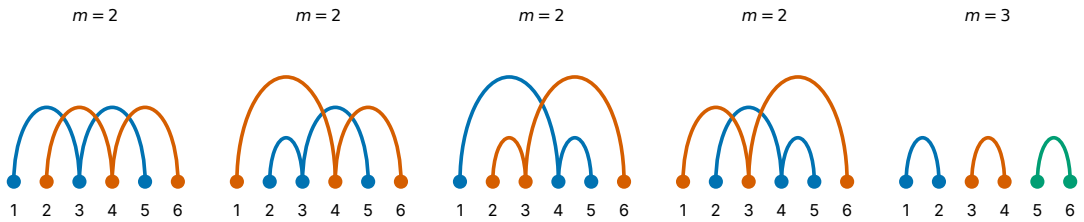


Figure 8: The five partitions of the all-ones word of the minimal reduced box with $k = 4$, $p = 6$ (representative $X = \{8, \dots, 13\} \subset \mathrm{AG}(3,2)$): four partitions into two circuits of size 3, one into three circuits of size 2. *Description:* exact data from the circuit list; blocks drawn as arcs over the six coordinates.

Proof. Each $V_P \setminus \{0, \mathbf{1}\}$ has $2^{m_P} - 2$ elements and these sets partition $C \setminus \{0, \mathbf{1}\}$ (Theorem 11.3), which has $2^k - 2$ elements. □

Corollary 12.2. *(a)* $n_2 \le 2^{k-1} - 1$, *with equality iff all partitions of* $\mathbf{1}$ *have exactly two blocks. (b)* $\beta = |\Pi| \le 2^{k-1} - 1$. *(c) If* $n_m = 0$ *for all* $m \ge 4$ *then* $2n_2 + 6n_3 = 2^k - 2$.

Proof. Immediate from $2^m - 2 \ge 2$ for $m \ge 2$ and Theorem 12.1. □

Example 12.3 (Checks)**.** E_4 ($k = 3$): $3 \cdot (4 - 2) = 6 = 2^3 - 2$. The minimal reduced box with $k = 4, p = 6$: $4 \cdot 2 + 1 \cdot 6 = 14 = 2^4 - 2$. $\mathrm{AG}(3,2)$ ($k = 4$, $p = 8$): $7 \cdot 2 = 14$. The classes with $k = 5$, $p = 10$ and $n_2 = 12$, $n_3 = 1$: $12 \cdot 2 + 6 = 30 = 2^5 - 2$. All 231 minimal reduced classes with $k \le 6$ satisfy the identity (exact check; Appendix L).

13 The second-moment (Parseval) identity

For a word w write $|w| = \mathrm{wt}(w)$.

Lemma 13.1 (Second moment of the weight)**.** *Let C be a reduced code of nullity k and length p (no dead coordinates, no two series-equivalent coordinates). Then*

$$\sum_{w \in C} |w|^2 = 2^{k-2}\, p\,(p+1).$$

Proof. $|w|^2 = \sum_{i,j} w_i w_j$. For each coordinate i, $\#\{w : w_i = 1\} = 2^{k-1}$ because the ith coordinate functional is nonzero on C. For $i \ne j$ the pair of functionals $(w \mapsto w_i, w \mapsto w_j)$ on C is linearly independent (the coordinates are neither dead nor series-equivalent: a relation $w_i = w_j$ for all w or $w_i = 0$ for all w is excluded), so $\#\{w : w_i = w_j = 1\} = 2^{k-2}$. Therefore $\sum_w |w|^2 = p\,2^{k-1} + p(p-1)2^{k-2} = 2^{k-2}p(p+1)$. □

(In coding-theoretic language the hypothesis says the dual code has minimum distance at least 3, i.e. C is an orthogonal array of strength 2; the formula is the second moment in the MacWilliams framework [7].)

Lemma 13.2 (Block expansion)**.** *Let P be a partition of $[p]$ into blocks of sizes $b_1, \dots, b_m$ and put $S_P = \sum_j b_j^2$. Then*

$$\sum_{w \in V_P} |w|^2 = 2^{m-2}\big(S_P + p^2\big).$$

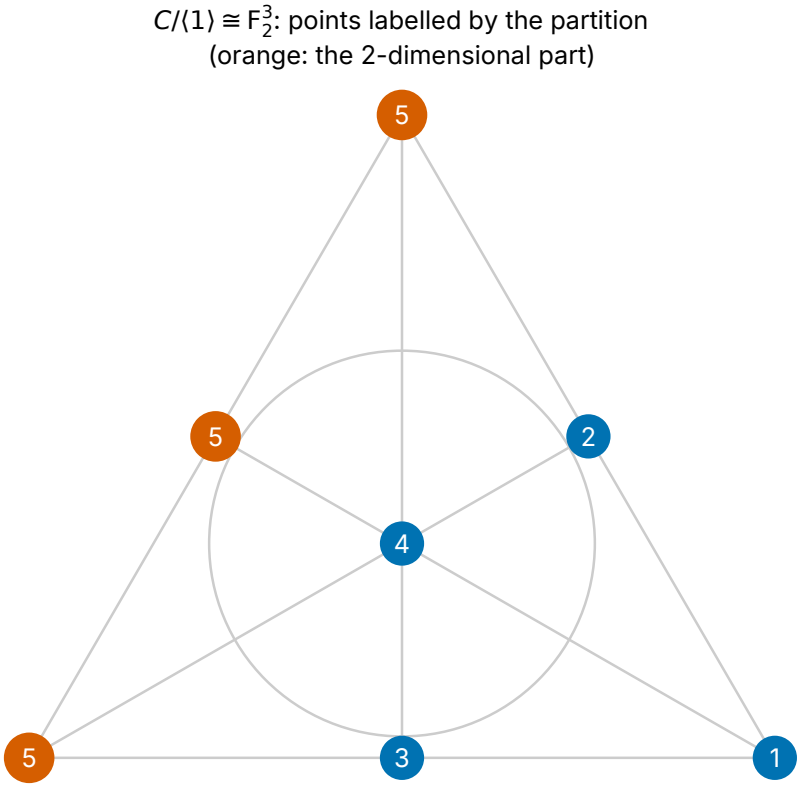


Figure 9: The vector-space partition of $C/\langle \mathbf{1}\rangle \cong \mathbb{F}_2^3$ (the Fano plane) for the same box: four singleton parts (points) and one part of dimension 2 (a line). *Description:* points are the seven nonzero cosets of $\langle \mathbf{1}\rangle$, computed exactly; drawing is schematic.

Proof. A word of V_P is the union of the blocks indexed by $J \subseteq [m]$, with weight $\sum_{j\in J} b_j$. Summing over all J: each b_j^2 appears in 2^{m-1} subsets, each cross term $b_j b_l$ ($j \neq l$, counted twice in the square) in 2^{m-2}, so $\sum_J (\sum_{j\in J} b_j)^2 = 2^{m-1} S_P + 2^{m-2}(p^2 - S_P) = 2^{m-2}(S_P + p^2)$. □

Theorem 13.3 (Parseval identity; `G11-L5b`)**.** *For a minimal non-embedded box (reduced, length p, nullity k),*

$$\sum_{P\in\Pi} \left(2^{m_P} S_P - 2p^2\right) = p\,(2^k - 2p),$$

where $S_P = \sum_{B\in P} |B|^2$. For a partition $P = \{B_1, B_2\}$ into two blocks the summand is $2(|B_1| - |B_2|)^2 \geq 0$.

Proof. Partition $C = \{0\} \sqcup \{\mathbf{1}\} \sqcup \bigsqcup_P (V_P \setminus \{0, \mathbf{1}\})$ (Theorem 11.3). The word 0 has weight 0 and $\mathbf{1}$ has weight p. By Lemmas 13.1 and 13.2,

$$2^{k-2} p(p+1) = p^2 + \sum_P \left[2^{m_P - 2}(S_P + p^2) - p^2\right].$$

Multiply by 4 and use $\sum_P 2^{m_P} = 2^k - 2 + 2\beta$ (Theorem 12.1, with $\beta = |\Pi|$):

$$2^k p(p+1) = 4p^2 + \sum_P 2^{m_P} S_P + p^2(2^k - 2 + 2\beta) - 4\beta p^2,$$

that is, $\sum_P 2^{m_P} S_P = 2^k p(p+1) - 4p^2 - p^2(2^k - 2) + 2\beta p^2 = 2^k p - 2p^2 + 2\beta p^2$. Subtracting $2\beta p^2$ gives $\sum_P (2^{m_P} S_P - 2p^2) = 2^k p - 2p^2 = p(2^k - 2p)$. For $m = 2$, $4(b_1^2 + b_2^2) - 2(b_1 + b_2)^2 = 2(b_1 - b_2)^2$. □

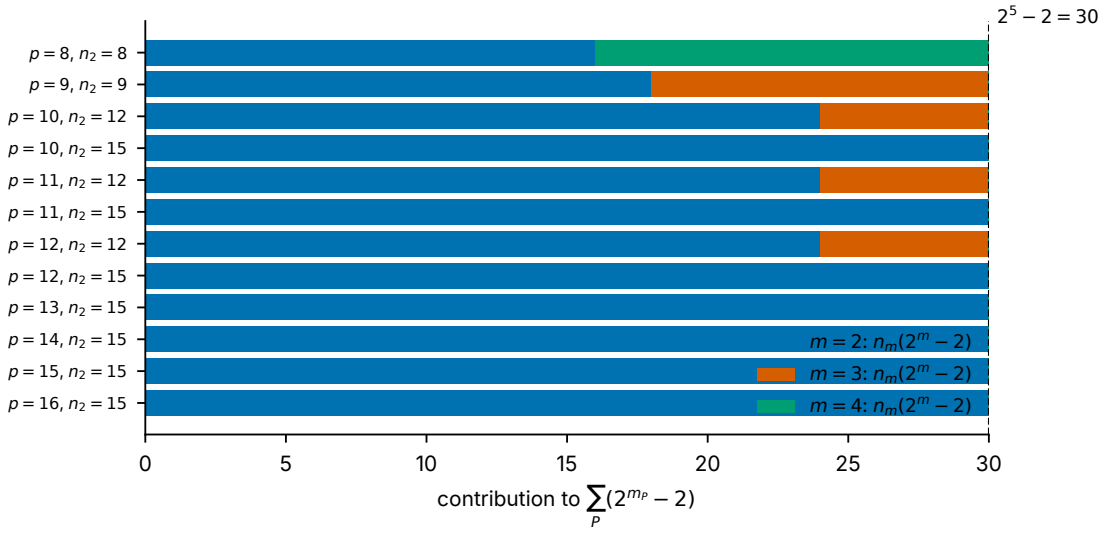


Figure 10: The counting identity for the twelve minimal reduced classes with $k = 5$: stacked contributions $n_m(2^m - 2)$ to the total $2^k - 2 = 30$. *Description:* exact from the classification of Section 19.

Source status. `G11-L5b`, GEO-I v011 `MATHEMATICS.md` §1.4 (PROVED). Re-derived here from Lemmas 13.1–13.2; the identity was verified exactly for all 231 classified classes with $k \le 6$ (Appendix L).

Example 13.4 (Checks)**.** E_4: $p = 4$, $k = 3$, three partitions with blocks $(2,2)$, so $S_P = 8$ and each summand is $4 \cdot 8 - 32 = 0$; the right side is $4\,(8 - 8) = 0$. For the $k = 4$, $p = 6$ box: the three-block partition has $S = 12$ and summand $8 \cdot 12 - 72 = 24$, the right side is $6\,(16 - 12) = 24$, so the four two-block summands $2(b_1 - b_2)^2$ sum to 0 and each two-block partition has two blocks of size 3. AG$(k-1, 2)$: all $2^{k-1} - 1$ partitions have blocks of size $p/2$, every summand vanishes, and the right side $p(2^k - 2p) = 0$ because $p = 2^{k-1}$.

Remark 13.5 (Fourier-analytic reading). The weight enumerator of C is the Fourier transform (over the group $\mathbb{F}_2^k$ of functionals) of the indicator of the point set X, and Lemma 13.1 is its second moment [13, 9]. The theorem says that this second moment is also the sum of the second moments of the pieces V_P of the vector-space partition, each of which is computable from the block sizes alone. This is what makes the partition data so rigid.

14 Quantitative inequalities

Put, for $P \in \Pi$ with $m = m_P$ blocks,

$$t_P = 2^m S_P - 2p^2 - p\,(2^m - 2).$$

Subtracting p times the counting identity from the Parseval identity gives

$$\sum_{P \in \Pi} t_P = p(2^k - 2p) - p(2^k - 2) = -2p(p-1). \tag{1}$$

Lemma 14.1 (Cauchy–Schwarz)**.** $S_P \ge p^2 / m_P$, *with equality iff all blocks of* P *have the same size* p/m_P.

Lemma 14.2 (Block bound)**.** $2 \le m_P \le p/2$ *for every* $P \in \Pi$.

Proof. Every block has size ≥ 2 (Corollary 7.3) and the sizes add to p. □

Lemma 14.3 (Two-block terms)**.** *If $m_P = 2$ then $t_P = 2(|B_1| - |B_2|)^2 - 2p \geq -2p$, with equality iff $|B_1| = |B_2|$.*

Lemma 14.4 (Threshold inequality)**.** *For integers $p \geq 9$ and $3 \leq m \leq p/2$,*

$$p\Big(\frac{2^m}{m} - 2\Big) \;\geq\; 2^m - 2,$$

with equality if and only if $(p, m) = (9, 3)$. For $p \in \{6, 7, 8\}$ the inequality fails for $m = 3$, and it holds strictly for $m = 4$, $p = 8$.

Proof. Since $2^m/m - 2 > 0$ for $m \geq 3$ the inequality is equivalent to $p \geq f(m) := \dfrac{m(2^m - 2)}{2^m - 2m}$ (the denominator is positive for $m \geq 3$). We compute $f(3) = \frac{3 \cdot 6}{2} = 9$, $f(4) = \frac{4 \cdot 14}{8} = 7$, $f(5) = \frac{5 \cdot 30}{22} = \frac{75}{11} < 7$. For $m = 3$ the inequality $p \geq 9$ holds with equality iff $p = 9$. For $m = 4$ we have $p \geq 8 > 7$ since $m \leq p/2$ gives $p \geq 8$. For $m \geq 5$ write $f(m) = m + \dfrac{m(2m - 2)}{2^m - 2m}$; then $f(m) < 2m$ is equivalent to $m(2m - 2) < m(2^m - 2m)$, i.e. $2^m > 4m - 2$, which holds for $m \geq 4$ ($16 > 14$) and propagates by $2^{m+1} = 2 \cdot 2^m > 8m - 4 \geq 4(m + 1) - 2$ for $m \geq 1$. Since $p \geq 2m > f(m)$ the inequality is strict. For $p \in \{6, 7, 8\}$ and $m = 3$: $p < 9 = f(3)$, so the inequality fails. □

m	$f(m)$	$\lceil f(m) \rceil$	$2m$	range of p where $p \geq f(m)$ and $p \geq 2m$
3	9	9	6	$p \geq 9$
4	7	7	8	$p \geq 8$
5	$75/11 \approx 6.82$	7	10	$p \geq 10$
6	$6 \cdot 62/52 \approx 7.15$	8	12	$p \geq 12$
7	$7 \cdot 126/114 \approx 7.74$	8	14	$p \geq 14$
8	$8 \cdot 254/240 \approx 8.47$	9	16	$p \geq 16$

Table 1: The threshold function $f(m) = m(2^m - 2)/(2^m - 2m)$. For every $m \geq 4$ one has $f(m) < 2m$, so the constraint $m \leq p/2$ already forces $p > f(m)$. Only $m = 3$ imposes a nontrivial lower bound, $p \geq 9$.

Corollary 14.5 (Sign of the higher terms)**.** *If $p \geq 9$ then $t_P \geq 0$ for every $P \in \Pi$ with $m_P \geq 3$, and $t_P > 0$ unless $(p, m_P) = (9, 3)$ and all three blocks have size 3.*

Proof. By Lemma 14.1, $t_P \geq p\big[p(2^m/m - 2) - (2^m - 2)\big]$ with $m = m_P \leq p/2$; apply Lemma 14.4. Equality forces $(p, m) = (9, 3)$ and equality in Cauchy–Schwarz. □

Remark 14.6 (The exceptional finite range). For $p \leq 8$ the bound of Corollary 14.5 fails only for $m = 3$ (which requires $p \geq 6$), and only for $p = 6, 7, 8$. There the Cauchy–Schwarz bound gives $t_P \geq p\big[\frac{2}{3}p - 6\big]$, i.e. -12, $-\frac{28}{3}$ and $-\frac{16}{3}$ for $p = 6, 7, 8$ (Appendix I). With such negative terms permitted, the identities and inequalities alone do not exclude a profile with at most one two-block partition: the supplementary scan `G11-V1` of the source reports, for $k = 5$, one numerically feasible profile with $p = 6$ which no code realizes (our enumeration in Table 2 finds no minimal box at $(k, p) = (5, 6)$). The cases $p \leq 8$ are therefore treated by exhaustive enumeration (Section 16).

15 At least $p-1$ two-block partitions

Theorem 15.1 ($n_2 \geq p-1$; `G11-L6`)**.** *Let C be a minimal non-embedded box (reduced, length $p \geq 9$, nullity k). Then*

$$n_2 \;\geq\; p-1.$$

If $p \geq 10$ then equality holds if and only if $X = \mathrm{AG}(k-1,2)$ (so $p = 2^{k-1}$ and $C = \mathrm{RM}(1,k-1)$). If $p = 9$ equality is impossible, so $n_2 \geq 9$.

Proof. By (1), $-2p(p-1) = \sum_P t_P$. Split into two-block and higher partitions. Lemma 14.3 gives $\sum_{m_P=2} t_P \geq -2p\, n_2$ and Corollary 14.5 gives $\sum_{m_P \geq 3} t_P \geq 0$. Hence $-2p(p-1) \geq -2pn_2$, i.e. $n_2 \geq p-1$.

Equality. If $n_2 = p-1$ all inequalities are tight: every two-block partition is balanced (blocks of size $p/2$) and $\sum_{m_P \geq 3} t_P = 0$. For $p \geq 10$ Corollary 14.5 gives $t_P > 0$ for every $m_P \geq 3$, so there is no partition with $m_P \geq 3$. Then the counting identity reads $2n_2 = 2^k - 2$, so $p-1 = n_2 = 2^{k-1}-1$ and $p = 2^{k-1} = |H|$: $X = H = \mathrm{AG}(k-1,2)$. Conversely $X = \mathrm{AG}(k-1,2)$ has $n_2 = p-1$ by Proposition 15.2. For $p = 9$, equality would force every $m_P \geq 3$ partition to have $m_P = 3$ with blocks $3,3,3$ (the exceptional case of Corollary 14.5) and so $2 \cdot 8 + 6n_3 = 2^k - 2$, i.e. $2^k = 18 + 6n_3$ is divisible by 3, which is absurd. □

Source status. `G11-L6`, GEO-I v011 `MATHEMATICS.md` §1.5 (PROVED+CHECKED); sharpness `G11-K1` (CHECKEDin the source). The characterisation of the equality case for $p \geq 10$ and the impossibility for $p = 9$ are made explicit here (new here). The bound was confirmed on all 231 classified classes and on 19 373 labelled minimal subsets for $k = 5$ (min $n_2 = 8$).

Proposition 15.2 (The Reed–Muller family)**.** *For $k \geq 3$ let $X = \mathrm{AG}(k-1,2)$ (all $p = 2^{k-1}$ points of H) and $C = \mathrm{RM}(1,k-1)$. Then the circuits are the $2^k - 2$ words of weight $p/2$; the partitions of $\mathbf{1}$ are the $2^{k-1}-1$ pairs $\{w, \mathbf{1} \oplus w\}$; $n_2 = p-1$; C is minimal non-embedded; $r_{\min} = 3$.*

Proof. The words are the affine functions on $\mathbb{F}_2^{k-1}$: 0, $\mathbf{1}$ and $2^k - 2$ words of weight $p/2$. Distinct nonzero words of equal weight are not nested, and every word other than $0, \mathbf{1}$ is nonconstant and has weight $p/2$, so the circuits are exactly the $2^k - 2$ words of weight $p/2$ (the word $\mathbf{1}$ contains each, so is not a circuit). Two disjoint circuits have total weight p, hence are complementary: the partitions of $\mathbf{1}$ are the pairs $\{w, \mathbf{1} \oplus w\}$, of which there are $(2^k - 2)/2$. For $k \geq 3$ this is ≥ 3, so C is non-embedded by Theorem 5.2. Shortening at a coordinate i: the circuits avoiding i are the w with $w_i = 0$; two disjoint ones are complementary, so one of them contains i: contradiction. Hence no set is partitioned into two or more circuits in $C_{(i)}$, every set has at most one partition, and $C_{(i)}$ is embedded. Rank three follows from Theorem 9.2. □

Source status. `G8-T3c`, GEO-I v008 `MINIMAL_RANK.md` (PROVED, CHECKEDfor $k = 3,4,5$); re-proved above by a direct argument.

16 W2 universality

The inequality of Section 15 settles $p \geq 9$. For $p \leq 8$ we use a normal form that reduces the problem to a very small enumeration.

Lemma 16.1 (Simplex normal form)**.** *Let $X \subseteq H \cong \mathrm{AG}(k-1,2)$ be a spanning subset with $|X| = p$. Then X is $\mathrm{AGL}(k-1,2)$-equivalent to a set*

$$\{e_1, \ldots, e_k\} \cup R, \qquad R \subseteq O_k,\ |R| = p-k,$$

where, in suitable coordinates, H is the set of odd-weight vectors of $\mathbb{F}_2^k$ and O_k is the set of odd-weight vectors of weight ≥ 3.

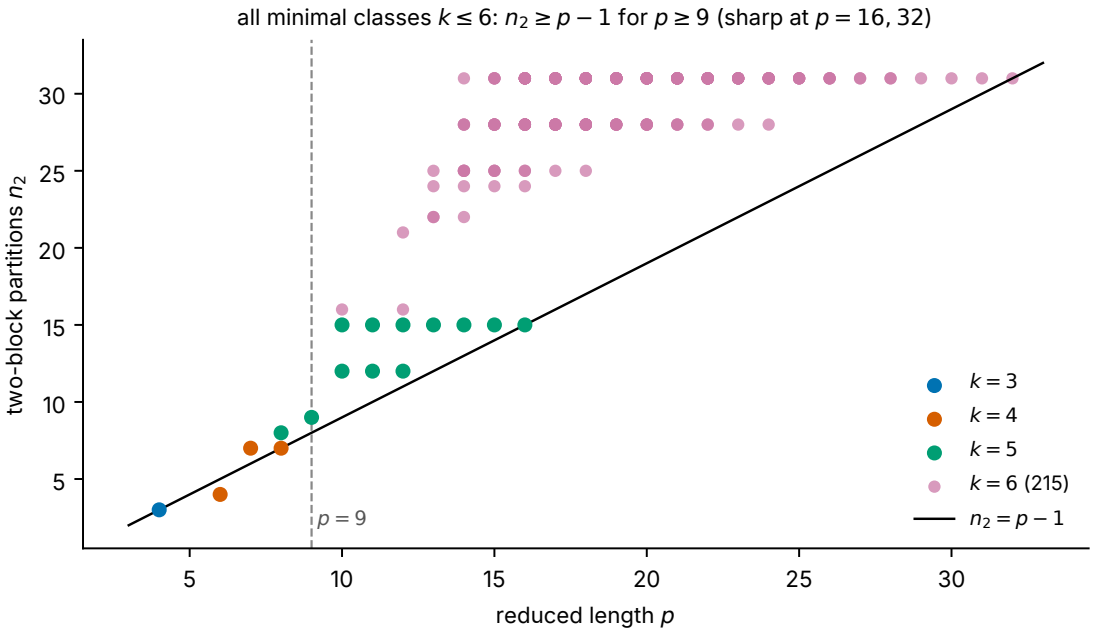


Figure 11: The number n_2 of two-block partitions against the reduced length p for all minimal classes with $k = 3, \dots, 6$ (231 classes), with the line $n_2 = p - 1$ and the threshold $p = 9$. Points on the line are the Reed–Muller supports $p = 4, 8, 16, 32$. *Description:* exact data; for $k = 6$ from the source-reported list of 215 representatives recomputed with the independent engine.

Proof. X spans $\mathbb{F}_2^k$, so it contains a basis $b_1, \dots, b_k$. Since $f_0(b_i) = 1$ for all i, in the basis (b_i) the functional f_0 is $(1, \dots, 1)$, so H is the set of odd-weight vectors and $b_i = e_i$. The change of basis fixes f_0, hence induces an element of $\mathrm{AGL}(k-1, 2)$ (Section 8). The remaining points of X are odd-weight vectors different from e_i, i.e. of weight ≥ 3. □

Proposition 16.2 (Finite check for $p \le 8$)**.** *For every pair (k, p) with $3 \le k < p \le 8$ let $\mathcal{N}_{k,p}$ be the set of subsets $R \subseteq O_k$ with $|R| = p - k$ (so $\binom{|O_k|}{p-k}$ candidates). For each candidate the reduced box $\{e_1, \dots, e_k\} \cup R$ was tested for non-embeddedness (Theorem 5.2, direct enumeration of circuit partitions), minimality (non-embeddedness of every shortening), and n_2 (the number of partitions of* **1** *into two circuits). The results are in Table 2: minimal boxes occur only for $(k, p) \in \{(3,4), (4,6), (4,7), (4,8), (5,8)\}$ and they have $n_2 \ge 3$.*

Source status. New here as a self-contained route (Appendix K); `scripts/small_p.py`, own code. The source proves W2 for $p \le 8$ via the classification through nullity 6 (`G11-T3`, `MATHEMATICS.md` §1.7) and the exclusion of $(k, p) = (7, 8)$ (`G11-L6b`). Both routes agree. Status of Proposition 16.2: PROVED+CHECKED(exhaustive, complete, 655 candidates in total; no hand proof).

Theorem 16.3 (W2 universality; `G11-T3`)**.** *Every minimal non-embedded box satisfies W2: the all-ones word has at least two partitions into exactly two circuits, i.e. $n_2 \ge 2$. More precisely, if $p \ge 9$ then $n_2 \ge p - 1 \ge 8$, and if $p \le 8$ then $n_2 \ge 3$.*

Proof. If $p \ge 9$, Theorem 15.1. If $p \le 8$, the reduced code is a spanning subset of $\mathrm{AG}(k-1, 2)$ with $k + 1 \le p \le 8$ points (Theorem 8.2), hence equivalent to one of the candidates of Proposition 16.2 by Lemma 16.1; by Table 2 every minimal one has $n_2 \ge 3$. The cases with $p = k + 1$ are consistent with this: they are the even-weight codes E_p, minimal only for $p = 4$ (Proposition 8.3); in particular the case $(k, p) = (7, 8)$ ($C = E_8$, with 28 circuits of weight two, 105 perfect matchings and $n_2 = 0$) is not minimal. □

p	k	candidates	non-embedded	minimal	min n_2 over minimal
4	3	1	1	1	3
5	3	0	0	0	—
5	4	4	4	0	—
6	3	0	0	0	—
6	4	6	6	6	4
6	5	11	11	0	—
7	3	0	0	0	—
7	4	4	4	4	7
7	5	55	55	0	—
7	6	26	26	0	—
8	3	0	0	0	—
8	4	1	1	1	7
8	5	165	165	10	8
8	6	325	325	0	—
8	7	57	57	0	—

Table 2: The finite enumeration for $p \leq 8$ (Proposition 16.2). "Candidates" counts the sets $R \subseteq O_k$ of size $p - k$. All candidates are non-embedded. Minimal boxes with $p \leq 8$ exist only for the pairs $(k, p) = (3, 4), (4, 6), (4, 7), (4, 8), (5, 8)$; for $(k, p) = (5, 8)$ ten of the 165 candidates are minimal.

Source status. `G11-T3` (= `G10-C1` = `G11-C2`), GEO-I v011 `MATHEMATICS.md` §1.7 (PROVED+CHECKED). The part for $p \geq 9$ is proved in the text without any enumeration. The part for $p \leq 8$ rests on the exhaustive enumeration above (and, in the source, on the classification for $k \leq 6$ and `G11-L6b`).

17 The main theorem: universal rank three

Theorem 17.1 (Universal minimum witness rank; `G11-T4`)**.** *For every non-embedded box C of the frozen binary $0/1$ Graver-cell model, of arbitrary length n, nullity k and parity matrix (zero columns included),*

$$r_{\min}(C) = 3.$$

The value 3 is attained by a collision of two cells of dimension 2 obtained from two two-circuit partitions of **1** *in a minimal sub-box.*

Proof. Theorem 16.3 shows that every minimal non-embedded box satisfies W2. Theorem 9.4 (descent to a minimal sub-box, Theorem 9.2 for the upper bound, Lemma 9.1 for the lower bound) yields $r_{\min}(C) = 3$. □

Source status. `G11-T4`, GEO-I v011 `RANK_UNIVERSAL.md` (PROVED). All steps are proved in this paper except the finite enumeration of Proposition 16.2.

Corollary 17.2 (No minimal box of rank four)**.** *No minimal non-embedded box has $r_{\min} \geq 4$. Consequently the conjecture `G10-C2` ("a minimal box with $r_{\min} = 4$ exists") is false in the frozen model, and the statement "$r_{\min} \geq 4$ implies nullity ≥ 7" (`G10-T4`) is vacuous.*

Source status. `G11-C3`: status FALSE for `G10-C2`, GEO-I v011 `RANK_UNIVERSAL.md`.

Remark 17.3 (What the theorem does not say). (i) Individual collisions can have large rank (Remark 4.9). (ii) The theorem is about the *existence* of a rank-three collision in every non-embedded box; it gives no bound on how hard it is to *find* one. (iii) The theorem is for the frozen binary model only. (iv) The proof of W2 for $p \leq 8$ is by a complete but computer-executed enumeration; the proof for $p \geq 9$ is a hand proof.

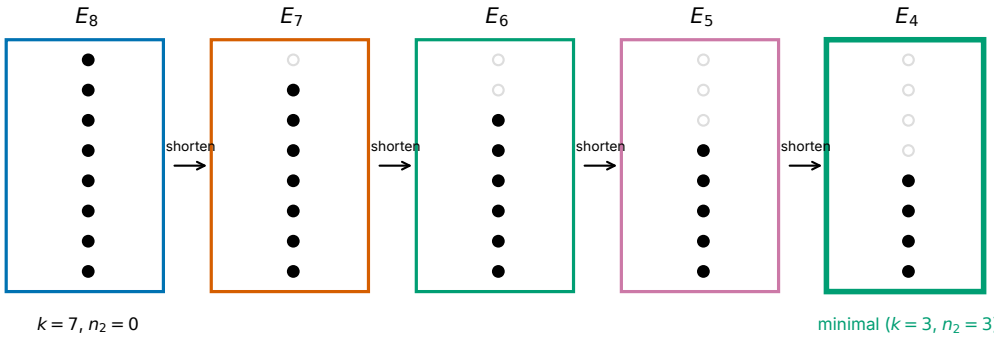


Figure 12: The case $(k, p) = (7, 8)$: $C = E_8$ with its 28 weight-two circuits and 105 perfect matchings of the coordinates, but no partition of **1** into two circuits. E_8 is not minimal because shortening gives E_7, then E_6, E_5, E_4, which is the minimal sub-box on four coordinates. *Description:* schematic chain of shortenings; the sub-box E_4 sits on any four of the eight coordinates.

Part IV
Examples, classification and consequences

18 The explicit rank-three obstruction

18.1 The even-weight code of length four

Let $C = E_4 = \{w \in \mathbb{F}_2^4 : \text{wt}(w) \text{ even}\}$, the kernel of $M = (1\,1\,1\,1)$; $k = 3$. We write words by their supports, e.g. 13 for $\{1, 3\}$.

Circuits. The six weight-two words $12, 13, 14, 23, 24, 34$ (Proposition 8.3); 1234 is not a circuit. Pairs of disjoint circuits: $\{12, 34\}$, $\{13, 24\}$, $\{14, 23\}$. So $\Pi = \{P_1, P_2, P_3\}$ with $P_1 = \{12, 34\}$, $P_2 = \{13, 24\}$, $P_3 = \{14, 23\}$, and $n_2 = 3$, $m_P = 2$ for all three. The counting identity reads $3 \cdot (2^2 - 2) = 6 = 2^3 - 2$.

Cells. The fibre is $F = \{12, 13, 14, 23, 24, 34, 1234\}$ (seven points). A direct enumeration from Definition 4.1 gives 7 vertices, 18 edges (6 circuits $\times$ 3 admissible cosets) and exactly 3 two-dimensional cells:

$$\begin{aligned} Z_A &= Z(\{12, 34\}, 13) \quad \text{vertices } 13,\ 14,\ 23,\ 24, \\ Z_B &= Z(\{13, 24\}, 12) \quad \text{vertices } 12,\ 14,\ 23,\ 34, \\ Z_C &= Z(\{14, 23\}, 12) \quad \text{vertices } 12,\ 13,\ 24,\ 34. \end{aligned}$$

Their unions are all $\{1, 2, 3, 4\}$, there is nothing outside the union, and the three centres are $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ (Theorem 4.4). Hence all three pairs collide. The cells Z_A, Z_B share the diagonal with endpoints 14 and 23; Z_A, Z_C share $13, 24$; Z_B, Z_C share $12, 34$. The cell structure is the three equatorial squares of the octahedron with vertices $12, 13, 14, 23, 24, 34$, together with the vertex 1234 and edges to it (Figure 7).

The rank. With the common base point $x = 14$ (not in $\operatorname{span} P_1 = \{0, 12, 34, 1234\}$ nor in $\operatorname{span} P_2 = \{0, 13, 24, 1234\}$), $s = (1 - 2x_i)_i = (-1, 1, 1, -1)$ and

$$g_1 = (-1, 1, 0, 0),\ g_2 = (0, 0, 1, -1); \qquad h_1 = (-1, 0, 1, 0),\ h_2 = (0, 1, 0, -1).$$

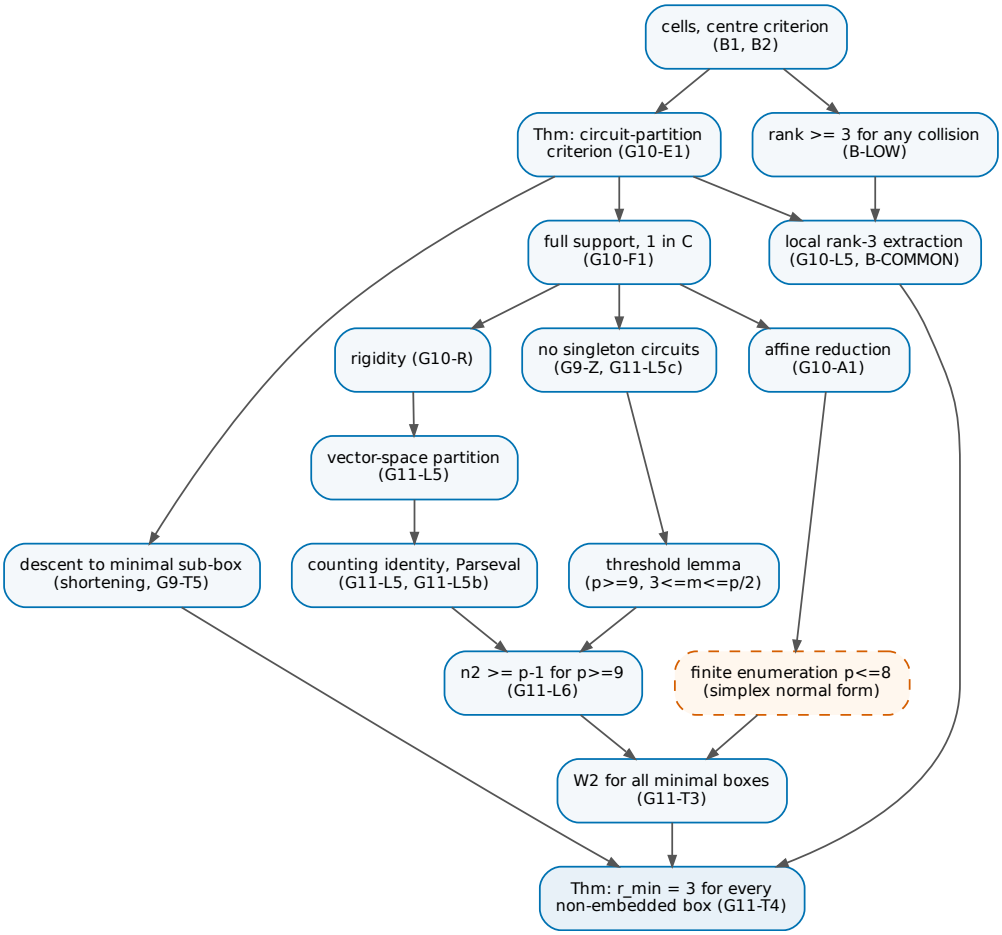


Figure 13: Dependency diagram of the proof. Arrows point from a result to the results that use it. Dashed boxes mark the finite enumeration ($p \leq 8$) and the supplementary source-reported material (nullity-6 classification), which is not used in the proof of the main theorem. *Description:* generated by `scripts/figs.py`.

Then $g_1 + g_2 = (-1, 1, 1, -1) = h_1 + h_2$. The vectors g_1, g_2, h_1 are independent: a relation $ag_1 + bg_2 + ch_1 = 0$ gives from coordinate 2 that $a = 0$, from coordinate 4 that $b = 0$ and then $c = 0$. So the witness rank is exactly 3. Because $r = t = 2$, the witness space has dimension $3 = r + t - 1$. By Lemma 9.1 no collision of E_4 has smaller rank; by Theorem 17.1 the same is true of every non-embedded box.

Remark 18.1 (Geometry of the witness). The three planes in $\mathbb{R}^4$ spanned by Z_A, Z_B, Z_C lie in the common affine hyperplane $\sum x_i = 2$, which is 3-dimensional. In that $\mathbb{R}^3$ the octahedron is a genuine picture: two equatorial squares meet along a diagonal and a third square passes through the common centre. The collision is thus a "crossing of two planes in a 3-space", the simplest possible crossing.

18.2 Minimality and uniqueness

Proposition 18.2. *E_4 is minimal (Proposition 8.3). Every minimal non-embedded box of nullity 3 is a blow-up of E_4 (that is, $p = 4$ and $X = \mathrm{AG}(2, 2)$).*

Proof. For $k = 3$ the affine space $H = \mathrm{AG}(2,2)$ has four points, so $p \le 4$, and $p \ge k+1 = 4$ gives $X = H$; the code is $\mathrm{RM}(1,2) = E_4$. □

18.3 Higher nullity

Reed–Muller family. By Proposition 15.2, for every $k \ge 3$ the box $\mathrm{RM}(1, k-1)$ of length 2^{k-1} is minimal non-embedded, with $2^{k-1} - 1$ two-circuit partitions of $\mathbf{1}$ into complementary pairs of weight $p/2$. For $k = 4$ ($n = 8$): circuits are the 14 words of weight 4; the rank-three collision uses any two of the seven partitions. For $k = 3$ it is E_4.

Blow-ups. Replacing coordinate i of E_4 by $\mu_i \ge 1$ copies gives minimal boxes of every length $n \ge 4$ and nullity 3 (Proposition 6.7); all have $r_{\min} = 3$.

Beyond Reed–Muller. The minimal reduced box with $k = 4$, $p = 6$ has point set (coordinates $1, \dots, 6$)

$$X = \{1000, 1001, 1010, 1011, 1100, 1101\} \subset \mathrm{AG}(3,2),$$

11 circuits, and the five partitions of $\mathbf{1}$ shown in Figure 8: the four two-block partitions $\{135|246\}$, $\{235|146\}$, $\{145|236\}$, $\{245|136\}$ and the three-block partition $\{12|34|56\}$. The identity $4 \cdot 2 + 6 = 14 = 2^4 - 2$ holds, as does Parseval's identity (Example 13.4). A rank-three collision comes from any two of the four two-block partitions. The class with $k = 5$, $p = 8$ has two-block partitions (eight of them) and one four-block partition $\{12|34|56|78\}$, with $8 \cdot 2 + 14 = 30$.

Remark 18.3 (Individual versus minimum, again). Exact enumeration shows that the $k = 4$, $p = 6$ box has 78 collisions (66 between cells of dimensions $(2,2)$ and 12 between $(2,3)$), of witness rank 3 (30 times) and 4 (48 times); the $k = 5$, $p = 8$ box has 1596 collisions (1540 of type $(2,2)$ and 56 of type $(2,4)$) with ranks $3, 4, 5$ occurring $364, 1176, 56$ times. The theorem concerns the *minimum*.

19 Exact finite classification

This section reports the classification of minimal reduced boxes for small nullity. It is *not used* in the proof of Theorem 17.1; its role is to document the examples, to provide independent evidence for the general theorems, and to record what is known beyond them.

19.1 What is classified

By Theorem 8.2 and Proposition 6.7, a minimal non-embedded box is determined, up to coordinate permutation and basis change, by (i) its nullity k, (ii) a spanning subset $X \subseteq \mathrm{AG}(k-1,2)$ up to $\mathrm{AGL}(k-1,2)$, and (iii) a multiplicity function $\mu : X \to \mathbb{Z}_{\ge 1}$. So classifying minimal boxes of nullity k means classifying the $\mathrm{AGL}(k-1,2)$-orbits of subsets X whose reduced box is minimal non-embedded; call these *minimal classes*.

19.2 Results

Theorem 19.1 (Classification for $k \le 5$; `G8-T3b`, `G9-T4`)**.** *The minimal reduced boxes with $k = 3, 4, 5$ form 1, 3 and 12 classes respectively, listed in Table 4. Each has $n_2 \ge p - 1$ whenever $p \ge 9$ and $n_2 \ge 3$ always; each satisfies the counting and Parseval identities; all have $r_{\min} = 3$.*

k	minimal classes	p range	status and provenance
3	1	4	PROVED+CHECKED; re-enumerated here
4	3	6, 7, 8	PROVED+CHECKED; re-enumerated here (37 labelled subsets)
5	12	8, . . . , 16	PROVED+CHECKED; re-enumerated here (19 373 labelled subsets)
6	215	10, . . . , 32	source-reported (`G10-T4`); the 215 representatives were re-checked here; completeness not re-derived

Table 3: Minimal classes of reduced boxes by nullity. "Re-enumerated" means that all $2^{2^{k-1}}$ subsets of $\mathrm{AG}(k-1,2)$ were examined with the independent engine of Appendix L (no use of the release code).

Proof. The proof is an exhaustive enumeration. For each $k \in \{3,4,5\}$ every subset $X \subseteq \mathrm{AG}(k-1,2)$ with $|X| \geq k+1$ that spans $\mathbb{F}_2^k$ (1, 93 and 57 541 subsets) was tested: non-embeddedness by Theorem 5.2 (enumeration of circuit partitions) and minimality by Definition 6.3 (shortening at each of the p coordinates). The minimal subsets (1, 37, 19 373) were grouped into $\mathrm{AGL}(k-1,2)$-orbits by closing under the generators of $\mathrm{AGL}(k-1,2)$ (translations and elementary transvections); the orbit counts are 1, 3, 12, and the orbit sizes 1; 28, 8, 1; 840, 4480, 6720, 448, 1680, 2688, 140, 1680, 560, 120, 16, 1 add up to the numbers of subsets above. □

Source status. `G8-T3b` (nullity 3, 4, v008 `MINIMAL_RANK.md`) and `G9-T4` (nullity 5, v009 `MATHEMATICS.md` §2.2), both PROVED+CHECKEDin the source. The enumeration here is independent of the release code; counts agree. The classification for nullity 5 in the source lists $p = 8, 9, 10, 10, 11, 11, 12, 12, 13, 14, 15, 16$; so does Table 4.

k	p	n_2	profile m^{n_m}	orbit size	|Stab|
3	4	3	2^3	1	24
4	6	4	$2^4, 3^1$	28	48
4	7	7	2^7	8	168
4	8	7	2^7	1	1344
5	8	8	$2^8, 4^1$	840	384
5	9	9	$2^9, 3^2$	4480	72
5	10	12	$2^{12}, 3^1$	6720	48
5	10	15	2^{15}	448	720
5	11	12	$2^{12}, 3^1$	1680	192
5	11	15	2^{15}	2688	120
5	12	12	$2^{12}, 3^1$	140	2304
5	12	15	2^{15}	1680	192
5	13	15	2^{15}	560	576
5	14	15	2^{15}	120	2688
5	15	15	2^{15}	16	20160
5	16	15	2^{15}	1	322560

Table 4: The 16 minimal classes of nullity 3, 4, 5. "Profile" lists the number n_m of partitions of $\mathbf{1}$ with m blocks. "Orbit size" is the number of labelled subsets of $\mathrm{AG}(k-1,2)$ in the class; $|\mathrm{Stab}| = |\mathrm{AGL}(k-1,2)|/\text{orbit size}$ (24, 1344, 322 560 for $k = 3, 4, 5$). Exact output of `scripts/classify.py`.

Burnside counts. The number of $\mathrm{AGL}(m,2)$-orbits on all subsets of $\mathrm{AG}(m,2)$ (including $\emptyset$) is $5, 10, 32$ for $m = 2, 3, 4$ (group orders $24, 1344, 322\,560$; computed here by Burnside's

lemma with exhaustively generated groups) and 382 for $m = 5$ (source-reported). For $k = m + 1 = 6$ this gives the source's chain of counts: 381 non-empty orbits, 350 spanning, 349 non-embedded, 215 minimal.

Theorem 19.2 (Nullity six; `G10-T4`, source-reported)**.** *There are* 215 *minimal classes of reduced boxes with* $k = 6$*, with* $10 \le p \le 32$ *and* $16 \le n_2 \le 31$*. Every class has W2, rank* 3*, satisfies the identities, and* $n_2 \ge p - 1$ *when* $p \ge 9$ *holds in all cases.*

Source status. `G10-T4`, GEO-I v010 `BINARY6.md` (PROVED+CHECKED: certificates `C6_rank6.json` with the checker `checker10.py`). *Independent verification here:* the list of 215 representative supports was read from the certificate file and, with the independent engine, each was confirmed to be a minimal non-embedded reduced box, to satisfy the counting and Parseval identities and $n_2 \ge p - 1$ ($p \ge 9$), and to have the recorded n_2; the 215 supports are pairwise inequivalent under $\mathrm{AGL}(5, 2)$ (separated by invariants, with an exhaustive backtracking search for the 8 pairs sharing invariants). *Not re-derived:* that the list is complete. Distribution of (p, n_2) in Table 5.

p	classes	$\min n_2$	$\max n_2$
10	1	16	16
12	2	16	21
13	4	22	25
14	10	22	31
15	15	24	31
16	25	24	31
17	23	25	31
18	26	25	31
19	22	28	31
20	22	28	31
21	16	28	31
22	15	28	31
23	9	28	31
24	8	28	31
25	5	31	31
26	4	31	31
27	2	31	31
28	2	31	31
29	1	31	31
30	1	31	31
31	1	31	31
32	1	31	31

Table 5: Nullity six: the 215 source-reported classes, recomputed with the independent engine (`scripts/verify6.py`).

19.3 Algorithms and certificates

Enumeration algorithm ($k \le 5$). For every subset X of $\mathrm{AG}(k-1, 2)$ (as a bitmask of the 2^{k-1} points): (1) discard it unless it has rank k and $p \ge k + 1$; (2) compute all $2^k - 1$ nonzero words and their circuits (inclusion-minimal supports); (3) enumerate all families of pairwise disjoint circuits and record which unions have two partitions (Theorem 5.2); (4) for each coordinate i repeat (3) on the circuits avoiding i (Lemma 6.2); (5) X is minimal iff (3) finds a double partition and (4) never does. Orbits: union–find over the generators of $\mathrm{AGL}(k-1, 2)$.

Certificates. A *partition certificate* for a point set $X \subseteq \mathrm{AG}(k-1, 2)$ consists of the points and two functionals $f \ne g$ whose words are disjoint circuits covering X, together with a

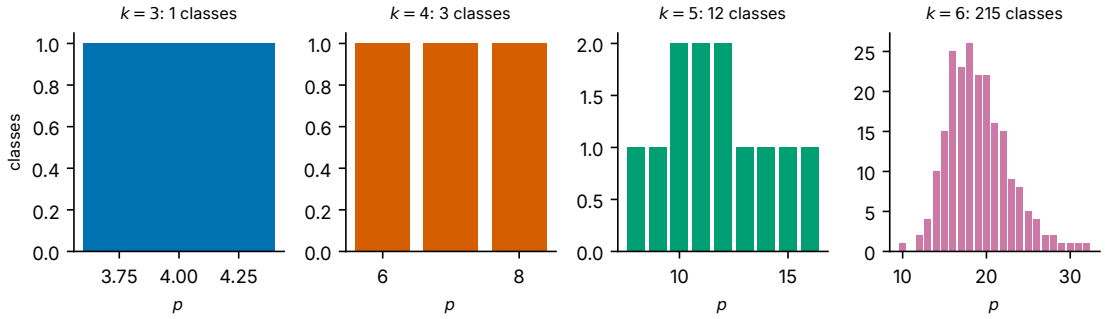


Figure 14: Classification summary for $k = 3, \ldots, 6$: number of minimal classes by reduced length p. *Description:* exact counts; $k = 6$ from the source-reported list recomputed here.

second such pair; a standalone checker confirms it using only rank computations (source format PARTITION, `checker11.py`). By Theorems 5.2 and 9.2 a valid certificate proves non-embeddedness with a rank-three collision. It does *not* certify minimality. All 231 source certificates (every minimal class with $k \leq 6$) were accepted by the unchanged source checker and, independently, by the engine here.

Separation of roles. The classification is a table of examples. The universal theorems in Part III do not depend on it: W2 for $p \geq 9$ is proved by inequality, W2 for $p \leq 8$ by a different small enumeration (Section 16), and Theorem 17.1 follows.

19.4 Historical statuses

Several statements recorded in the release series were later settled; their current status is recorded here for completeness (details in Appendix M): `G8-C3` (full support of minimal boxes) was open in v008–v009 and is proved as Theorem 7.1; `G10-C1` (universal $r_{\min} = 3$) was open in v010 and is Theorem 17.1; `G10-C2` (existence of a rank-four minimal box) is FALSE (Corollary 17.2); the v010 implication "$r_{\min} \geq 4$ requires nullity ≥ 7" is vacuous.

Question 1 (Spanning sets). Is the box of every spanning subset $X \subseteq \mathrm{AG}(k-1, 2)$ with $|X| \geq k+1$ non-embedded? This holds for $k \leq 5$ in the exhaustive enumeration here, for $k = 6$ in the source counts (350 spanning orbits, 349 non-embedded; the one exceptional orbit is presumably the simplex $p = k$, for which $C = \mathbb{F}_2^k$ is embedded, but we have not checked this against the source enumeration), and in all of 540 random samples with $k \in \{6, 7, 8\}$, $p - k \in \{2, 3, 4\}$ tested here. It is not proved.

20 Implications

20.1 Uniform obstruction dimension

Theorem 17.1 says that the *obstruction to embeddedness has a uniform local size*: every non-embedded box, of any length n and nullity k, contains a pair of 2-dimensional cells in a common 3-dimensional affine subspace whose relative interiors cross at their centres. In geometric language: non-embedding never needs a configuration of higher dimension than the crossing of two planes in 3-space. The ambient dimension n may be arbitrarily large, and span of all cells may be n-dimensional, but the obstruction is already visible in dimension three.

Corollary 20.1 (A short certificate)**.** *A box C is non-embedded if and only if it contains a word a with two distinct decompositions $a = b_1 \sqcup b_2 = c_1 \sqcup c_2$ into two disjoint circuits.*

Proof. ($\Leftarrow$) Theorem 9.2. ($\Rightarrow$) By Proposition 6.4 and Theorem 16.3 there is a minimal subbox $C_{(I)}$ with two two-circuit partitions of its all-ones word; their circuits are circuits of C (Lemma 6.2), and $a = \mathbf{1}_{[n]\setminus I}$ is a word of C with two decompositions. □

20.2 Rigidity, finite affine structure, partition universality

Rigidity. In a minimal box the circuit structure is determined by a single failure of unique decomposition (Theorem 10.1); the number of cells at the centre is $\sum_P (2^{k-m_P} - 1)$ (Proposition 5.4); for example E_4 has 3, the $k = 4$, $p = 6$ box has 13, $\mathrm{AG}(3, 2)$ has 21, the $k = 5$ classes have $57, 69, 87, 105$ for $p = 8, 9, 10, 10$ (verified by direct enumeration).

Finite affine structure. Minimal boxes are exactly (up to blow-up) the subsets $X \subseteq \mathrm{AG}(k-1, 2)$ whose code has the rigidity property of Proposition 10.2. The problem of classifying all minimal boxes becomes the problem of classifying certain subsets of affine spaces over $\mathbb{F}_2$ up to the affine group.

Partition universality. Every minimal box carries a vector-space partition of $\mathbb{F}_2^{k-1}$ built from its circuit partitions, satisfying the counting identity and the Parseval identity; at least $p - 1$ of its parts (for $p \geq 9$) are one-dimensional.

20.3 Local witness versus global complexity

Remark 20.2 (What rank three does and does not give). Corollary 20.1 gives a certificate of non-embeddedness of size $O(n)$ (five subsets of $[n]$): the verification that b_1, b_2, c_1, c_2 are circuits of C summing to the same word requires only rank computations of column submatrices of M (a set T is a circuit iff the columns of M indexed by T have a one-dimensional kernel spanned by $\mathbf{1}_T$). So non-embeddedness has polynomially checkable certificates in the binary-matrix encoding. This does **not** yield a polynomial-time *algorithm*: the theorem asserts the existence of such a word but not how to find it, and the number of candidates for a is exponential in n. We make no claim about the computational complexity of deciding embeddedness; and "bounded witness rank" must not be read as "efficient detection". The size of the local witness is uniformly 3; the cost of locating it is unaddressed.

20.4 Consequences for the arithmetic model

Within the arithmetic setting of the GEO-I series the binary boxes arise as the $0/1$ slices of congruence lattices. The theorem implies that, whenever a binary relation lattice (for a given set of active primes and parity configuration) has an exceptional cell overlap, that overlap already occurs in a 3-dimensional exponent subspace and is generated by exactly two pairs of 2-dimensional moves. Whether a particular arithmetic fibre is of this binary type is a separate question (Remark 2.4), which this paper does not address.

Part V
Relation to established mathematics, provenance, and open questions

21 Relations to established mathematics

This section compares the objects and results of the paper with standard notions and theorems in neighbouring areas. It is a comparison of definitions and statements, not a survey of the literature, and *no claim of novelty or priority is made*.

What was read, and how. The following sources were read in full text (as retrieved on the date of the manuscript): Heden's survey of vector-space partitions [3]; Kurz's paper on Heden's tail bound [15] (for the exact statement of Heden's theorem); Kurz's classification of vector-space partitions of $\mathrm{GF}(2)^8$ [16]; Cooper–Okur on multigraphs with a unique partition into cycles [17]; Margolis–Saliola–Steinberg (for the statement and axioms of the Bohne–Dress theorem on zonotopal tilings) [18]; and D. S. Kim's paper [19] (for the statement of the Pless power-moment identities, which it attributes to MacWilliams–Sloane). The remaining references (MacWilliams–Sloane, Huffman–Pless, Oxley, Graver, Sturmfels, Schrijver, Ziegler, Matoušek, Hirschfeld, Beutelspacher, Terras, O'Donnell, Hemmecke) were assembled from catalogue records and author or publisher pages and were *not* read; statements attributed to them are general background to be checked by the reader and no theorem number from them is quoted. Where a theorem of a read source is used below, it is stated as it appears there, and any consequence drawn is marked as a consequence derived here and not as a statement of the source.

21.1 Binary codes

The code $C = \ker M$ is an ordinary binary linear code [7, 6]. A *circuit* in our sense is a word of C whose support is minimal under inclusion among supports of nonzero words; in coding theory such words are usually called *minimal codewords*. Lemma 3.2(b) is the familiar fact that every codeword is a disjoint union of minimal codewords (a fact that is special to the binary case). The reduced code, in which no two coordinates are series-equivalent and none is dead, is the code with no repeated or zero columns in its generator matrix; equivalently the dual code has minimum distance at least 3, i.e. C is an orthogonal array of strength 2, and Lemma 13.1 is the corresponding second moment of the weight distribution. The family $\mathrm{RM}(1, k-1)$ of Proposition 15.2 is the first-order Reed–Muller code, a classical code with weights $0, p/2, p$. The even-weight code E_p (parity-check code) is the simplest example of a code whose minimal codewords are pairs.

We did not find, in the sources read, the study of the *set of partitions of the all-ones word into minimal codewords* as an object in its own right (the object of Part III), nor the specific link between such partitions and vector-space partitions of $C/\langle \mathbf{1} \rangle$ (Theorem 11.3); the subsection on vector-space partitions below records precisely what *is* classical in that part. This is an observation about the sources read, not a statement about the literature as a whole.

21.2 Matroids and graphs

The circuits of C are the circuits of the binary matroid represented by the parity matrix M (its column matroid), and equally the cocircuits of the vector matroid represented by a generator matrix; C is the cocycle space of the latter [10]. The decomposition of a binary cocycle (or cycle) into disjoint circuits is standard for binary matroids; non-uniqueness of that decomposition is the phenomenon behind Theorem 5.2.

Example 21.1 (Cut spaces of graphs)**.** If C is the cut space of a connected graph G (the set of edge cuts $\delta(S)$, as words in $\mathbb{F}_2^{E(G)}$), the circuits of C are the *bonds* (minimal nonempty cuts), and Theorem 5.2 reads: the box is non-embedded if and only if some cut has two distinct decompositions into edge-disjoint bonds. The even-weight code E_p is the cut space of the cycle graph C_p (bonds are pairs of edges), and the cut $\delta(\{1,3\}) = \{12,23,34,41\}$ of C_4 has the three decompositions $\{12,34\}$, $\{23,41\}$ and so on; in this language E_4 is the smallest obstruction. The $K_{2,t}$ cut spaces mentioned in the release series (`G10-K`, CHECKED for $t \leq 10$ in the source) belong to this family; we do not treat them here.

Remark 21.2 (Cycle codes and the theorem of Cooper and Okur)*.* For the *cycle space* of a (multi)graph G, the circuits are the cycles, the codewords are the Eulerian subgraphs, and Theorem 5.2 says: the box is non-embedded iff some Eulerian subgraph has two different partitions of its edge set into cycles. Cooper and Okur [17] (Corollary 3.1 for multigraphs, Theorem 2.8 for digraphs, in the numbering of the arXiv version read) prove that a connected Eulerian multigraph has a *unique* partition into cycles if and only if it is a bridgeless cactus multigraph, i.e. no two of its cycles share an edge. Combining the two statements (a translation made here; their paper does not use cycle spaces, matroids or codes): *the box of the cycle code of a multigraph G is embedded if and only if every connected Eulerian subgraph of G is a cactus.* The smallest obstruction is two vertices joined by four parallel edges, whose cycle code is E_4 and whose three partitions into digons are the three partitions of E_4 in Section 18. This graphic specialisation of Theorem 5.2 is thus a known theorem of graph theory, and our criterion extends it to arbitrary binary matroids. As a consistency check of the translation (not a proof), the equivalence between the criterion of Theorem 5.2 and "some connected Eulerian subgraph contains two cycles sharing an edge" was tested exhaustively on 2000 random multigraphs with at most 5 vertices and 9 edges, with no discrepancy (`scripts/lit_checks.py`). For cut spaces of planar graphs the duality (bonds of G are cycles of the dual) transfers the same statement to bond partitions; we have not examined which of the minimal classes of Part IV are graphic or cographic.

21.3 Finite affine geometry

A spanning subset $X \subseteq \mathbb{F}_2^k$ determines the code $C(X)$, and the circuits of $C(X)$ are complements of hyperplanes of X.

Lemma 21.3 (Circuits as hyperplane complements)**.** *For $f \in \mathbb{F}_2^k \setminus \{0\}$ the word w_f is a circuit of $C(X)$ if and only if $X \cap f^\perp$ spans the hyperplane $f^\perp$ (that is, has rank $k-1$).*

Proof. For $g \neq 0$, $\operatorname{supp} w_g \subseteq \operatorname{supp} w_f$ iff $X \cap f^\perp \subseteq X \cap g^\perp$. If $X \cap f^\perp$ has rank $k-1$ this forces $g^\perp \supseteq f^\perp$, so $g = f$. If the rank is smaller, choose $g \notin \{0, f\}$ vanishing on $\operatorname{span}(X \cap f^\perp)$: then $w_g \neq w_f$ is a nonzero word with support inside that of w_f, hence strictly smaller (the supports cannot coincide because $f \mapsto w_f$ is injective). □

This is the test used in the source checker and in our engine (Appendix L). Specialising to two-block partitions gives a geometric meaning for n_2.

Proposition 21.4 (Two-block partitions as parallel hyperplane pairs)**.** *Let $X \subseteq H \cong \mathrm{AG}(k-1,2)$ be a spanning subset and φ a non-constant affine function on H. Then $\{X \cap \{\varphi = 1\},\ X \cap \{\varphi = 0\}\}$ is a partition of* **1** *into two circuits of $C(X)$ if and only if $X \cap \{\varphi = 0\}$ and $X \cap \{\varphi = 1\}$ each* affinely span *the affine hyperplane $\{\varphi = 0\}$, respectively $\{\varphi = 1\}$, of H. Consequently*

$$n_2 = \#\{\textit{directions of parallel hyperplane pairs of } \mathrm{AG}(k-1,2) \\ \textit{both of whose sides are affinely spanned by } X\} \ \le\ 2^{k-1} - 1.$$

Proof. Write f_1 for the linear functional with $f_1|_H = \varphi$; then $f_2 = f_1 + f_0$ satisfies $f_2|_H = 1 + \varphi$. The supports of w_{f_1}, w_{f_2} are $X \cap \{\varphi = 1\}$ and $X \cap \{\varphi = 0\}$ — disjoint and covering X. By Lemma 21.3 they are circuits iff $X \cap f_1^{\perp} = X \cap \{\varphi = 0\}$ spans $f_1^{\perp}$ and $X \cap f_2^{\perp} = X \cap \{\varphi = 1\}$ spans $f_2^{\perp}$. As $0 \notin H$, a subset of $H \cap f^{\perp}$ spans the linear hyperplane $f^{\perp}$ iff it affinely spans the affine hyperplane $H \cap f^{\perp}$. Directions of parallel pairs correspond to nonzero linear parts, of which there are $2^{k-1} - 1$; the pair $\{\varphi, 1 + \varphi\}$ describes one direction. The partition is determined by the direction. □

Source status. New here (a reformulation; no statement from the release files is replaced). The equality $n_2 =$ #directions was verified on all 231 classified classes (Appendix L).

Thus W2 asks that *two different directions of* $\mathrm{AG}(k-1,2)$ *cut X into two affinely spanning halves*. The extreme case is $X = \mathrm{AG}(k-1,2)$, for which every direction works ($n_2 = 2^{k-1} - 1 = p - 1$, Theorem 15.1).

21.4 Graver bases, lattice ideals and zonotopal tilings

Graver bases of integer lattices are classical in integer programming and algebraic statistics [2, 11, 12]. The relation lattice $\Lambda_M \supseteq 2\mathbb{Z}^n$ is a very special lattice for which the Graver basis is explicitly described by Proposition 3.6: sign vectors on binary circuits plus the vectors $\pm 2e_i$. A cell is a parallelepiped with Graver generators, i.e. a zonotope with independent generators [14]; unions of such cells, and the question of when they fit together face to face, resemble the theory of zonotopal tilings, in which tiles of a tiling of a zonotope meet face to face by definition. In the present setting, however, we consider the set of *all* admissible cells rather than a chosen tiling, and the question is whether this set is a complex; a collision is exactly a place where two cells of the family cannot belong to a common tiling in which they appear as tiles. Margolis, Saliola and Steinberg [18] state the Bohne–Dress theorem as a bijection between zonotopal tilings of the zonotope $Z(\mathcal{A})$ of a vector configuration and single-element liftings of the associated oriented matroid, with a tiling defined by four axioms: (ZT1) the tiles cover $Z(\mathcal{A})$; (ZT2) faces of tiles are tiles; (ZT3) two tiles meet in a common face or not at all; (ZT4) every edge of $Z(\mathcal{A})$ is a tile. Our face-to-face condition (Proposition 4.6) is the analogue of (ZT3). The remaining axioms have no counterpart in our setting: our cells lie in the cube $[0,1]^n$ and are not required to cover a zonotope or to include its edges, so the theorem is not applicable as stated and we draw no conclusion from the analogy. We do not pursue the connection.

21.5 Vector-space partitions

A *vector-space partition* of a finite vector space V is a collection of subspaces such that every nonzero vector lies in exactly one of them; it has *type* $[d_1^{n_1} \dots d_t^{n_t}]$ if it has n_i members of dimension d_i [3]. The set $\{V_\Pi / \langle \mathbf{1} \rangle\}$ of Theorem 11.3 is a vector-space partition of $C/\langle \mathbf{1} \rangle \cong \mathbb{F}_2^{k-1}$, with parts of dimension $m_\Pi - 1$; its 1-dimensional parts are the n_2 two-block partitions. What follows separates what is classical about this object from what is specific to the present setting.

Classical: the packing condition. Heden's survey calls the identity $\sum_i n_i(q^{d_i}-1) = q^n - 1$ the *packing condition* (Section 5.1; it is not numbered in the survey). In our quotient it reads $\sum_\Pi (2^{m_\Pi - 1} - 1) = 2^{k-1} - 1$, which is the counting identity of Theorem 12.1 divided by 2. So Theorem 12.1 is the classical packing condition; the content of Theorem 11.3 is that the partition arises from circuit partitions of $\mathbf{1}$, not the identity itself. The *dimension condition* $d_i + d_j \le n$ (attributed in the survey to Bu) reads $(m_\Pi - 1) + (m_{\Pi'} - 1) \le k - 1$ here and is already a consequence of $V_\Pi \cap V_{\Pi'} = \{0, \mathbf{1}\}$.

Classical: types of small vector-space partitions. According to the survey (Section 5.3), all types of vector-space partitions of $V(n,2)$ were determined for $n \le 7$ by El-Zanati and coauthors, and Kurz [16] completes GF(2)[8]. The partitions arising here for $k \le 6$ live in $\mathbb{F}_2^{k-1}$ with $k - 1 \le 5$, so their types fall in the range where the list of admissible types is known. We did *not* enumerate that list, and therefore did not compare the types of Appendix J with it beyond the necessary condition checked below; the comparison remains open.

Classical: Heden's tail theorem, and a consequence derived here. The *tail* of a vector-space partition is the set of its parts of smallest dimension. Heden's theorem on the length u_1 of the tail, in the form quoted by Kurz [15] (the source text of that statement is partly garbled; the cases below are as reconstructed there and in the survey): if the type is $d_l^{u_l} \dots d_2^{u_2} d_1^{u_1}$ with $u_1, u_2 > 0$, then (i) if $q^{d_2 - d_1} \nmid u_1$ and $d_2 < 2d_1$ then $u_1 \ge q^{d_1} + 1$; (ii) if $q^{d_2-d_1} \nmid u_1$ and $d_2 \ge 2d_1$ then either $d_1 \mid d_2$ and $u_1 = [d_2]_q / [d_1]_q$, or $u_1 > 2q^{d_2-d_1}$; (iii) if $q^{d_2-d_1} \mid u_1$ and $d_2 < 2d_1$ then $u_1 \ge q^{d_2} - q^{d_1} + q^{d_2-d_1}$; (iv) if $q^{d_2-d_1} \mid u_1$ and $d_2 \ge 2d_1$ then $u_1 \ge q^{d_2}$.

Proposition 21.5 (A consequence of Heden's theorem; derived here)**.** *Let C be a minimal non-embedded box with $n_2 \ge 1$ and with at least one partition of $\mathbf{1}$ having $m_\Pi \ge 3$ blocks, and let $d = \min\{m_\Pi : m_\Pi \ge 3\} - 1$ (≥ 2). If Heden's theorem holds as quoted above, then $n_2 \ge 2^d - 1 \ge 3$. In particular $n_2 \ge 2$ (the statement W2) holds in every such box.*

Proof. In the quotient F_2^{k-1} the partition has tail of dimension $d_1 = 1$ and length $u_1 = n_2$, and the next smallest dimension is $d_2 = d \ge 2$ with $u_2 > 0$. As $d_2 \ge 2 = 2d_1$ only (ii) and (iv) can apply, with $q = 2$. In case (iv), $u_1 \ge 2^{d_2}$. In case (ii), either $u_1 = [d_2]_2 / [1]_2 = 2^{d_2} - 1$ or $u_1 > 2 \cdot 2^{d_2 - 1} = 2^{d_2}$. In all cases $u_1 \ge 2^{d_2} - 1 \ge 3$. □

Source status. Derived here from the theorem as quoted in [15]; the theorem itself is not re-proved. The inequality $n_2 \ge 2^d - 1$ was checked on all 231 classified classes (87 of them have a part with $m_\Pi \ge 3$), with no violation (`scripts/lit_checks.py`). Status: DERIVED FROM A CITED THEOREM; not used anywhere in the proofs of this paper.

What the classical theory does not give. Proposition 21.5 *assumes* $n_2 \ge 1$. Nothing in the packing, dimension or tail conditions forces a 1-dimensional part to exist (a tail may have larger dimension), and nothing relates n_2 to the number p of reduced coordinates. The existence of a two-block partition (W2) and the bound $n_2 \ge p - 1$ for $p \ge 9$ with its equality cases (Theorem 15.1) use the weight function: the second-moment identity (Lemma 13.1) and the Parseval identity of Theorem 13.3, which have no counterpart in the survey as read. In the other direction, the bound $n_2 \ge p - 1$ is much stronger than the classical $n_2 \ge 2^d - 1$ when p is large relative to 2^d. With the caveat that we read only one survey and two papers on the subject, we found no statement in them that implies W2 for the partitions arising here.

Examples. For E_4 ($k = 3$) the quotient is $\mathbb{F}_2^2$ and the partition is $[1^3]$ (the three points). For $k = 4, p = 6$ (Appendix G) the quotient is $\mathbb{F}_2^3$ and the type is $[2^1\,1^4]$: a line and four points, in accordance with $u_1 = 4 \geq 2^2 - 1$. For $X = \mathrm{AG}(k-1, 2)$ the type is $[1^{2^{k-1}-1}]$, the partition of $\mathbb{F}_2^{k-1}$ into points.

21.6 Fourier analysis on $\mathbb{F}_2^k$

The weight enumerator of a binary linear code is determined by the Fourier transform of the indicator of its point set X [7, 13, 9]. The Pless power-moment identities (stated for example as Theorem 5.1 in [19], where they are attributed to MacWilliams and Sloane) express the moments $\sum_j j^h B_j$ of the weight distribution of a code through the weight distribution of its dual; for $h = 2$ and a dual of minimum distance at least 3 they give the closed form of Lemma 13.1. We proved Lemma 13.1 directly and did not verify the general identity beyond this reading. Lemma 13.1 is the second power moment; Theorem 13.3 decomposes it along the vector-space partition. A natural question is whether higher moments yield further constraints (Question 7).

21.7 Embedding obstructions

The word "embedding" is used literally: let K_{abs} be the abstract cell complex whose cells are the $Z(S, x)$, with the face relations of Proposition 4.6, and let $\iota : K_{\mathrm{abs}} \to \mathbb{R}^n$ be the realization. A point of K_{abs} lies in the relative interior of exactly one cell, so ι is injective iff the images of the relative interiors of distinct cells are disjoint iff there is no collision; thus the box is embedded iff ι is an embedding. Classical obstructions to embedding cell complexes in Euclidean space—for instance the van Kampen–Flores theorem [8]—are *topological* (they apply to the abstract complex regardless of realization). The obstruction here is of a different nature: K_{abs} always exists, and non-embeddedness is a property of one particular map, forced by a purely linear-algebraic feature of the code (Theorem 5.2). Whether K_{abs} could be embedded in some other way in $\mathbb{R}^n$ or in a higher-dimensional space is a separate question that we do not consider.

21.8 What the comparison suggests

The pieces that connect to standard areas are: minimal codewords and their disjoint decompositions (codes, matroids); hyperplane complements and affine subsets (finite geometry); vector-space partitions; Fourier analysis on $\mathbb{F}_2^k$. The pieces that are specific to the setting are the Graver cells, collisions and the witness rank, which have no direct standard counterpart that we know.

22 Provenance, verification and reproducibility

22.1 Tiers of evidence

T1. *Source-reported proof.* A proof recorded in the GEO-I releases (v006–v011; Section 1). Statements and, where available, proofs were read; none was independently verified by an outside party.

T2. *Internal written audit.* The re-derivation of the arguments for this manuscript by its authors (the model-assisted drafting process described in the acknowledgements). A second reading by the same hand; not an external review.

T3. *Independent computational tests.* Programs written afresh for this manuscript (Appendix L) that test finite instances: exact integer / bitmask arithmetic throughout (no floating point is used in the mathematical checks).

T4. *External review.* None has taken place.

T5. *Formal proof.* None.

Each universal theorem of Parts II–III is proved in the text (tiers T1+T2) with one exception: the case $p \leq 8$ of Theorem 16.3 rests on an exhaustive computer enumeration (Proposition 16.2, tier T3 in the sense of being a complete finite verification, with a different route available in the source).

22.2 Finite computations accompanying the paper

Computation	Range	Instances	Failures
E1 criterion vs. direct cell/collision enumeration; $r_{\min} = 3$	all multisets over $\mathbb{F}_2^2, \mathbb{F}_2^3 \setminus 0$, $n \leq 7$, plus random codes over $\mathbb{F}_2^4 \setminus 0$ with $n \in [5, 8]$ (up to 1500)	4213 codes (706 non-embedded)	0
Circuit test: minimal supports vs. hyperplane rank test	same codes	4213	0
Classification by exhaustive enumeration	$k = 3, 4, 5$	$1 + 93 + 57\,541$ spanning subsets	—
Orbit counts / Burnside	$\mathrm{AGL}(m, 2)$, $m = 2, 3, 4$	$24, 1344, 322\,560$ elements	—
Identities (counting, Parseval, vector-space partition, pairwise $V_P \cap V_Q = \{0, \mathbf{1}\}$)	all minimal classes $k \leq 6$	$16 + 215$ classes	0
Centre cell count (Proposition 5.4)	all 16 classes with $k \leq 5$	16	0
Two-block partitions = hyperplane directions (Proposition 21.4)	all 231 classes	231	0
Finite check for $p \leq 8$ (Proposition 16.2)	$3 \leq k < p \leq 8$	655 candidates	—
215 nullity-6 classes pairwise inequivalent	invariants; exhaustive search for 8 pairs	215	0 identifications
Heden tail bound $n_2 \geq 2^d - 1$ (Proposition 21.5)	all 231 classes	87 with a part $m_\Pi \geq 3$	0
Cycle codes of multigraphs: E1 vs. Cooper–Okur cactus criterion (Remark 21.2)	random multigraphs, ≤ 5 vertices, ≤ 9 edges	2000	0 mismatches
Source checkers (unchanged) on source certificate files	`checker11` (PARTITION), `checker10` (RANK6), `checker8` (RANK)	$231, 215, 12$	0
Random spanning subsets non-embedded (Question 1)	$k \in \{6, 7, 8\}$	540	0 embedded

Table 6: Finite computations performed for this manuscript. They are support for statements and examples, never proofs of universal claims (with the exception of the finite enumeration of Proposition 16.2, which is a complete verification of a finite statement).

22.3 Source-reported items not re-derived

The following were taken from the releases without independent derivation here: the completeness of the nullity-6 list (`G10-T4`: 381 orbits, 350 spanning, 349 non-embedded, 215 minimal); the Burnside count 382 for $m = 5$; the supplementary scans `G11-V1` (knapsack feasibility), the random nullity-7/8 tests (`v11_w2.json`: 154 and 2 boxes, no violation); and the statement `G10-K` for $K_{2,t}$ ($t \leq 10$).

22.4 Disclosures

1. The GEO-I releases v001–v012 are cited at the fixed commit `343f9aef2b6697466058165c5fc2ddf0f00e882b` of the repository. The drafting process could not itself fetch the repository and read the local release folders supplied by the author instead; the local copies were not compared byte for byte with the pinned commit. No frozen release and no repository file was modified.
2. Of the bibliography entries other than the GEO-I releases, six were read in full text for Section 21 ([3, 15, 16, 17, 18, 19]); the others were assembled from catalogue records or author and publisher pages and were not read. Section 21 states which is which.
3. The figures are generated by scripts in the package from exact data; captions were written by hand and checked against the data. Three-dimensional renderings are exact pictures of the 3-dimensional objects they show (the E_3 triangle, the octahedron in the hyperplane $\sum x_i = 2$) and are never presented as renderings of higher-dimensional geometry.
4. Theorem 17.1 depends on the enumeration of Proposition 16.2; all other steps are hand proofs.
5. Items marked "new here" (Appendix M) were derived in the course of writing this manuscript and were not found in the release files read; they have not been refereed.

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23 Open questions

Question 2 (Beyond the 0/1 box). What happens for boxes $Q > \mathbf{1}$ in binary relation lattices (cells with corners at coordinates up to Q_i), where Graver vectors $\pm 2e_i$ now matter? Is there an analogue of Theorem 5.2, and is $r_{\min}$ still uniformly bounded?

Question 3 (Other lattices and moduli). For congruence lattices modulo $q > 2$, or with several residue conditions, circuits are replaced by the (much larger) Graver basis. Is there an exact criterion for collisions of cells in terms of decompositions of lattice vectors into Graver vectors? For which lattices is the minimum witness rank uniformly bounded?

Question 4 (Topology versus embedding). The realized union $\iota(K_{\text{abs}})$ of a non-embedded box has homology that need not be that of K_{abs}. How do the Betti numbers of the image relate to the collisions? For E_4 the three squares sharing diagonals give a specific small complex; is there a general formula in terms of the partition data?

Question 5 (Collision structures). Classify, for a given box, the set of occurring witness ranks and cell-dimension pairs. For E_6 the pairs are $(2,2)$ and $(3,3)$ with ranks 3–6; what is the general pattern? Is the number of rank-three collisions determined by (k, p, type)?

Question 6 (Types of vector-space partitions). Which types of vector-space partitions of $\mathbb{F}_2^{k-1}$ arise as $\{\bar{V}_P\}$? In the classified range there are eight types for $k = 6$ (maximum block number 5). Is the number of blocks of a partition of $\mathbf{1}$ bounded in terms of k more strongly than by $m \leq p/2$? Can the minimal boxes with $n_2 = p - 1$ be classified for $p < 9$?

Question 7 (Higher moments). Do the third and fourth moments of the weight distribution, decomposed along the partition, give further constraints that prove structural statements (for instance an upper bound on m_P) without enumeration?

Question 8 (Algorithmic complexity). What is the complexity of deciding embeddedness of a given code? Corollary 20.1 puts the problem in NP in the binary-matrix encoding; is it in P, or NP-complete? Is it fixed-parameter tractable in the nullity k?

Question 9 (Spanning subsets). Is every spanning subset $X \subseteq \mathrm{AG}(k-1,2)$ with $|X| \geq k+1$ non-embedded (Question 1)? A positive answer would give a complete characterisation of non-embedded boxes with $\mathbf{1} \in C$ in finite-geometric terms.

Question 10 (Classification beyond $k = 6$). Re-derive the nullity-6 classification independently (for instance by the simplex normal form of Lemma 16.1 and orbit enumeration), and determine the number of minimal classes for $k = 7$.

Question 11 (Cut spaces). For which graphs is the cut space a minimal box? Is $K_{2,t}$ minimal for all $t \geq 3$ (checked for $t \leq 10$ in the source), and is its $r_{\min}$ explained by a family of two-bond partitions?

Question 12 (Formal verification). The proofs of Parts II–III are short and elementary; they are natural targets for formalization in a proof assistant, with the finite enumeration for $p \leq 8$ replaced by a verified computation.

Appendices

A Notation registry and reconciliation with Paper A

This appendix fixes every symbol used in the paper, and records how the notation differs from that of the accepted companion manuscript (Paper A). Nothing in Paper B depends on reading Paper A; the reconciliation is given only so that a reader of both can translate.

Symbol	Meaning
n	number of coordinates (length of the code); the box is $[0,1]^n$.
M, $C = \ker M$	binary matrix and its code $C \subseteq \mathbb{F}_2^n$; $k = \dim C$ is the *nullity*.
Λ_M	the lattice $\{\alpha \in \mathbb{Z}^n : M\alpha \equiv 0 \bmod 2\}$.
$F = C \setminus \{0\}$	the fibre: the integer points of the box in Λ_M other than the origin.
$\operatorname{supp} w$, $\operatorname{wt} w$	support and Hamming weight of w.
$\mathcal{C}(C)$	the set of circuits (minimal supports of nonzero words); words and supports are identified.
S, $U(S)$, $\operatorname{span} S$	a family of pairwise disjoint circuits, its union, and the $2^{\lvert S\rvert}$ words obtained by taking unions of members.
$Z(S,x)$	the cell with base point $x \in C \setminus \operatorname{span} S$ and generators $g_j = (1-2x_i)_{i\in S_j}$.
p, X	reduced length (number of series classes) and the reduced point set $X \subseteq \mathrm{AG}(k-1,2)$.
Π	a partition of $\mathbf{1}$ into circuits; $V_\Pi = \operatorname{span}\Pi$, $m_\Pi = \dim V_\Pi$.
n_m, n_2	number of partitions of $\mathbf{1}$ into exactly m circuits; n_2 counts the two-block partitions.
N	number of cells through the centre of the box (Proposition 5.4).
$r_{\min}$	minimum witness rank: the least $\dim \operatorname{aff}(Z \cup Z')$ over collisions (Z, Z').
E_p	the even-weight code of length p (cut space of the cycle C_p).

Differences from Paper A. (i) Paper A indexes coordinates by a number of primes that it also calls k; here n is the length and k is always the nullity. (ii) Paper A treats boxes attached to arbitrary parity data; Paper B restricts attention to the frozen binary model, and for the minimal non-embedded boxes the relevant parity vector is the all-ones word. (iii) Terminology: "witness rank" and "collision" mean exactly what they mean in Paper A and the GEO-I releases; $r_{\min}$ is the minimum over collisions and is not the ambient dimension, the nullity, or the rank of every witness. (iv) All theorem numbers of this paper are new; Appendix M maps them to release identifiers.

B The Graver basis of Λ_M: further detail

Proposition 3.6 states that the Graver basis of Λ_M consists of the ± 1 sign vectors supported on circuits together with the vectors $\pm 2e_i$ for coordinates i that are not singleton circuits. We record the correspondence with the standard theory [2, 12, 4] and the two points that are easy to misstate.

Conformal order. For $u, v \in \mathbb{Z}^n$ write $u \sqsubseteq v$ if $u_i v_i \geq 0$ and $|u_i| \leq |v_i|$ for all i. The Graver basis is the set of $\sqsubseteq$-minimal nonzero elements of Λ_M. An element of the box fibre is a $0/1$ vector $w \in C$; its sign-extensions $\pm w$ are the lattice vectors it represents.

Why only circuits and doubles appear. Let $\alpha \in \Lambda_M$ be $\sqsubseteq$-minimal. If some $|\alpha_i| \geq 3$ then $\alpha - 2\operatorname{sgn}(\alpha_i)e_i \in \Lambda_M$ is conformally below α, a contradiction; so $|\alpha_i| \leq 2$. If $|\alpha_i| = 2$ then $\alpha \sqsubseteq 2e_i$ forces $\alpha = \pm 2e_i$. Otherwise $\alpha \in \{0, \pm 1\}^n$, its support is the support of a nonzero codeword, and minimality of α among sign-compatible vectors is exactly minimality of the support, i.e. the support is a circuit. Finally $\pm 2e_i \in \Lambda_M$ is $\sqsubseteq$-minimal unless $\pm e_i \in \Lambda_M$, which happens exactly when $\{i\}$ is a circuit.

Pitfall. The doubles $\pm 2e_i$ are Graver elements but are *not* corners of any cell of the box: their endpoints leave $[0,1]^n$. They are irrelevant to the geometry of the box, and Proposition 3.7

uses only the circuit part. Statements that quantify over "all Graver elements" must therefore be restricted before being applied to the $0/1$ model.

C Cells, collisions and the centre count: formal definitions

Cells. For a family $S = \{S_1, \dots, S_r\}$ of disjoint circuits and $x \in C \setminus \operatorname{span} S$, the cell $Z(S, x)$ is the convex hull of the 2^r points $x + \sum_{j \in J} g_j$ with $J \subseteq [r]$ (with $g_j = (1 - 2x_i)_{i \in S_j}$ extended by zero). By Lemma 4.2 a cell is determined by the pair $(S,\ x + \operatorname{span} S)$; the vertices are the words of the coset $x + \operatorname{span} S$ (mod 2 the coset is closed under addition of members of $\operatorname{span} S$, and the $0/1$ representatives are exactly the vertices).

Collision. Two distinct cells of positive dimension *collide* if their relative interiors intersect (Lemma 4.3). A box is *embedded* iff no collision exists, equivalently (Proposition 4.6) iff any two cells meet in a common face (possibly empty).

Centre count. Every collision of a minimal non-embedded box is centred at $\frac{1}{2}\mathbf{1}$ (Theorem 4.4 together with Theorem 7.1). The cells whose relative interior contains the centre are in bijection with the pairs $(\Pi,\ x + V_\Pi)$ with $x \notin V_\Pi$; counting cosets gives $2^{k - m_\Pi} - 1$ cells for each partition Π. Summing yields $N = \sum_\Pi (2^{k-m_\Pi} - 1)$, and since all N cells contain the centre in their relative interior any two of them collide, so there are exactly $\binom{N}{2}$ collisions (Corollary 5.5). Example: E_4 has $k = 3$ and three partitions of $\mathbf{1}$ into two circuits, $\{12, 34\}, \{13, 24\}, \{14, 23\}$, each with $m = 2$; each gives $2^{3-2} - 1 = 1$ cell, so $N = 3$ and there are $\binom{3}{2} = 3$ collisions.

D The circuit-partition criterion: variants

The proof of Theorem 5.2 rests on Lemma 5.1: an $\mathbb{F}_2$-vector space is not the union of two proper subspaces. Three comments on its use.

Where the lemma enters. In (ii)⇒(i) of Theorem 5.2, Lemma 5.1 supplies a base point $x \notin \operatorname{span} S \cup \operatorname{span} T$; this is the only place where the field $\mathbb{F}_2$ is used, and it is why the two cells share a base point and hence satisfy the centre criterion.

The case $S = T$. In (i)⇒(ii) the collision may a priori consist of two cells over the *same* circuit family S with different cosets. The proof shows this cannot happen without a second partition of U: the difference $c = x \oplus y \in C \setminus \operatorname{span} S$ is supported in U, and decomposing c and $\mathbf{1}_U \oplus c$ into circuits gives a partition $S' \neq S$.

Collisions at a centre. By Theorem 4.4, collisions of a minimal box are exactly the pairs of distinct cells through the centre; their number is $\binom{N}{2}$ (Appendix C). The criterion therefore does more than detect non-embedding: it enumerates all collisions.

E Shortening, minimality and full support: proofs of the lemmas

Let $C_{(i)} = \{w|_{[n]\setminus\{i\}} : w \in C,\ w_i = 0\}$ be the shortening of Definition 6.1 (equivalently, the kernel of M with column i deleted). Lemma 6.2(a) says that the circuits of $C_{(i)}$ are exactly the circuits of C avoiding i. The proof is a direct comparison of supports: words of $C_{(i)}$ are the

words of C vanishing at i; any word of C with support contained in a circuit avoiding i also vanishes at i, so minimality transfers in both directions.

Descent (Proposition 6.4). If C is non-embedded and some single shortening is again non-embedded, replace C by it; the number of coordinates strictly decreases, so the process stops at a coordinatewise-minimal non-embedded box. Collisions in $C_{(i)}$ lift to collisions in C with the same affine-span dimension, hence the minimum witness rank of C is at most that of the minimal sub-box, and Part III bounds the latter.

Full support (Theorem 7.1). Let C be minimal and suppose a doubly-partitioned word U misses a coordinate i. Then both partitions are circuit partitions of U inside $C_{(i)}$ by Lemma 6.2, so $C_{(i)}$ is non-embedded by Theorem 5.2, contradicting minimality. Hence $U = [n]$, and $\mathbf{1} = U \in C$.

F Affine reduction and the classification method

Generator matrices. Choose a basis of C and let G be the $k \times n$ generator matrix; its columns are points of $\mathbb{F}_2^k$. Since $\mathbf{1} \in C$, there is a functional that takes the value 1 on every column, so the columns lie in an affine hyperplane $H \cong \mathrm{AG}(k-1,2)$ (Theorem 8.2). Identical columns form series classes and are merged (Proposition 6.7); the result is the set $X \subseteq H$ of p distinct points, spanning $\mathbb{F}_2^k$.

Equivalence. Two minimal boxes are equivalent if their reduced point sets differ by an element of $\mathrm{AGL}(k-1,2)$ (Lemma 8.1). The group has order $2^{k-1}\prod_{i=0}^{k-2}(2^{k-1}-2^i)$, namely 24, 1344, 322 560 for $k = 3, 4, 5$. For each $k \le 5$ `scripts/classify.py` enumerates all $2^{2^{k-1}}$ subsets of H, selects those that are spanning with every shortening embedded and the full box non-embedded, and groups them into orbits using union–find under generators of $\mathrm{AGL}(k-1,2)$; the orbit sizes divide the group order and the Burnside count $\sum|\mathrm{Fix}|/|G|$ confirms the number of orbits. The result is Theorem 19.1: 1, 3, 12 classes for $k = 3, 4, 5$.

Simplex normal form. For $p \le 8$ one may by an affine change of coordinates assume that X contains the standard simplex $\{e_1, \dots, e_k\}$ (Lemma 16.1); the remaining $p-k$ points are odd-weight vectors of weight ≥ 3, and the finite check of Proposition 16.2 runs over all such completions.

G Rigidity and the vector-space partition: a worked example

Take $k = 4$, $p = 6$ and the point set $X = \{1000, 1001, 1010, 1011, 1100, 1101\} \subseteq \mathrm{AG}(3,2)$ (columns of the generator matrix, written in binary). The code has $2^4 - 1 = 15$ nonzero words and eleven circuits:

$$12,\ 34,\ 56,\ 135,\ 136,\ 145,\ 146,\ 235,\ 236,\ 245,\ 246.$$

The partitions of $\mathbf{1} = 123456$ into circuits are

$$\{12, 34, 56\}, \qquad \{135, 246\},\ \{145, 236\},\ \{136, 245\},\ \{146, 235\},$$

i.e. $n_3 = 1$, $n_2 = 4$ (the profile $2^4, 3^1$ of Table 4). The subspaces V_Π have dimensions $m = 3$ for the three-block partition and $m = 2$ for each of the four others, and they pairwise intersect in $\{0, \mathbf{1}\}$ (Theorem 11.3).

Counting identity. $\sum_\Pi (2^{m_\Pi} - 2) = 4 \cdot 2 + 1 \cdot 6 = 14 = 2^4 - 2$.

Parseval identity. $\sum_\Pi (2^{m_\Pi} S_\Pi - 2p^2)$ with $S_\Pi = \sum_{c \in \Pi} |c|^2$: the four two-block partitions have $S = 9 + 9 = 18$ and contribute $4 \cdot 18 - 72 = 0$ each; the three-block partition has $S = 4 + 4 + 4 = 12$ and contributes $8 \cdot 12 - 72 = 24$. Total $24 = p(2^k - 2p) = 6 \cdot 4$. Both identities are thus checked by hand on this example; Table 6 records the exact verification for all classified classes.

Rigidity (Theorem 10.1). Every word other than $\mathbf{1}$ has a unique circuit decomposition: for instance $1234 = 12 \sqcup 34$ is the only decomposition of this word, and 135 is a circuit.

H Counting and Parseval: derivations

The derivations of Theorems 12.1 and 13.3 are given in full in Part III. We restate the chain of equalities in one place and list the exact identities verified on every classified class.

1. Partition $C \setminus \{0, \mathbf{1}\}$ into the sets $V_\Pi \setminus \{0, \mathbf{1}\}$ (Theorem 11.3); counting elements gives $\sum_\Pi (2^{m_\Pi} - 2) = 2^k - 2$.
2. Sum the squared weight over C: $\sum_{w \in C} \mathrm{wt}(w)^2 = 2^{k-2} p(p+1)$ for a reduced code with no repeated coordinate (Lemma 13.1).
3. Expand the same sum block by block (Lemma 13.2): grouping the words of $C \setminus \{0, \mathbf{1}\}$ by the block V_Π containing them and subtracting the common contribution of 0 and $\mathbf{1}$ yields $\sum_\Pi (2^{m_\Pi} S_\Pi - 2p^2) = p(2^k - 2p)$.

k	p	profile	counting	Parseval
3	4	2^3	$6 = 6$	$0 = 0$
4	6	$2^4, 3^1$	$14 = 14$	$24 = 24$
4	7	2^7	$14 = 14$	$14 = 14$
4	8	2^7	$14 = 14$	$0 = 0$
5	8	$2^8, 4^1$	$30 = 30$	$128 = 128$
5	9	$2^9, 3^2$	$30 = 30$	$126 = 126$
5	10	$2^{12}, 3^1$	$30 = 30$	$120 = 120$
5	10	2^{15}	$30 = 30$	$120 = 120$
5	11	$2^{12}, 3^1$	$30 = 30$	$110 = 110$
5	11	2^{15}	$30 = 30$	$110 = 110$
5	12	$2^{12}, 3^1$	$30 = 30$	$96 = 96$
5	12	2^{15}	$30 = 30$	$96 = 96$
5	13	2^{15}	$30 = 30$	$78 = 78$
5	14	2^{15}	$30 = 30$	$56 = 56$
5	15	2^{15}	$30 = 30$	$30 = 30$
5	16	2^{15}	$30 = 30$	$0 = 0$

Table 7: Exact values of the two sides of the counting and Parseval identities for the classes with $k \leq 5$ (output of `scripts/geo2.py::check_identities`). The identities are also verified for the 215 classes with $k = 6$ (Table 6).

I Inequalities and the finite exceptional range

Part III shows $n_2 \geq p-1$ for $p \geq 9$ from the threshold lemma $p(2^m/m-2) \geq 2^m-2$ for $p \geq 9$, $3 \leq m \leq p/2$. The three smallest cases of the function $f(m) = m(2^m-2)/(2^m-2m)$ are $f(3) = 9$, $f(4) = 7$, $f(5) = 75/11$ (Table 1), so only $m = 3$ is binding and it is binding exactly at $p = 9$.

The range $p \leq 8$. For $p \leq 8$ the threshold inequality does not exclude minimal boxes whose partitions have large blocks, and the universality of W2 is established instead by exhaustive enumeration in simplex normal form (Proposition 16.2, Table 2). The counts of candidate completions and of minimal boxes among them are reproduced from `scripts/small_p.py` and `logs/small_p.json`. Minimal boxes with $p \leq 8$ occur only for $(k,p) \in \{(3,4),(4,6),(4,7),(4,8),(5,8)\}$ and each has $n_2 \geq 2$. The code E_8 ($k = 7$, $p = 8$) is not minimal (Figure 12).

J Construction of rank-three witnesses

Let $\Pi = \{b_1, b_2\}$ and $\Pi' = \{c_1, c_2\}$ be two distinct partitions of $\mathbf{1}$ into two circuits (these exist in every minimal box, Theorem 16.3). Choose $x \in C \setminus (V_\Pi \cup V_{\Pi'})$, which exists by Lemma 5.1. The cells $Z(\Pi, x)$ and $Z(\Pi', x)$ are parallelograms that collide (Theorem 5.2). Their generators are the restrictions of the single vector $s = (1-2x_i)_{i\in[n]}$ to the four blocks, so $g_1 + g_2 = s = h_1 + h_2$ is a linear relation among four vectors and the rank is at most 3; Lemma 9.1 gives at least 3 (Theorem 9.2). In the table x is recorded as the set of coordinates carrying the value 1 in a chosen base point, and the rank is computed independently by exact Fraction elimination, for the first listed minimal class of each (k,p) with $k \leq 5$.

k	p	partition Π	partition Π'	base x	rank
3	4	$\{1,2\} \mid \{3,4\}$	$\{1,3\} \mid \{2,4\}$	$\{2,3\}$	3
4	6	$\{1,3,5\} \mid \{2,4,6\}$	$\{1,4,5\} \mid \{2,3,6\}$	$\{1,2\}$	3
4	7	$\{1,2,3,4\} \mid \{5,6,7\}$	$\{1,4,5\} \mid \{2,3,6,7\}$	$\{2,3,5\}$	3
4	8	$\{1,2,3,4\} \mid \{5,6,7,8\}$	$\{1,2,5,6\} \mid \{3,4,7,8\}$	$\{3,4,5,6\}$	3
5	8	$\{1,3,5,7\} \mid \{2,4,6,8\}$	$\{1,4,5,7\} \mid \{2,3,6,8\}$	$\{1,2\}$	3
5	9	$\{1,5,6,7\} \mid \{2,3,4,8,9\}$	$\{1,3,5,8\} \mid \{2,4,6,7,9\}$	$\{1,2,3\}$	3
5	10	$\{1,4,5,8\} \mid \{2,3,6,7,9,10\}$	$\{1,4,6,7,8\} \mid \{2,3,5,9,10\}$	$\{1,2,3,4\}$	3
5	10	$\{1,2,3,4,5,6\} \mid \{7,8,9,10\}$	$\{1,2,4,7\} \mid \{3,5,6,8,9,10\}$	$\{3,5,6,7\}$	3
5	11	$\{1,4,6,7,9\} \mid \{2,3,5,8,10,11\}$	$\{1,4,5,8,9\} \mid \{2,3,6,7,10,11\}$	$\{1,2,3,4\}$	3
5	11	$\{1,2,3,4,5,6,7\} \mid \{8,9,10,11\}$	$\{1,4,6,7,8\} \mid \{2,3,5,9,10,11\}$	$\{2,3,5,8\}$	3
5	12	$\{1,2,5,6,9,10\} \mid \{3,4,7,8,11,12\}$	$\{1,2,7,8,9,10\} \mid \{3,4,5,6,11,12\}$	$\{1,2,3,4\}$	3
5	12	$\{1,2,3,4,5,6,7,8\} \mid \{9,10,11,12\}$	$\{1,4,6,7,9\} \mid \{2,3,5,8,10,11,12\}$	$\{2,3,5,8,9\}$	3
5	13	$\{1,2,3,4,5,6,7,8\} \mid \{9,10,11,12,13\}$	$\{1,2,7,8,9,10\} \mid \{3,4,5,6,11,12,13\}$	$\{3,4,5,6,9,10\}$	3
5	14	$\{1,2,3,4,5,6,7,8\} \mid \{9,10,11,12,13,14\}$	$\{1,2,7,8,9,10\} \mid \{3,4,5,6,11,12,13,14\}$	$\{3,4,5,6,9,10\}$	3
5	15	$\{1,2,3,4,5,6,7,8\} \mid \{9,10,11,12,13,14,15\}$	$\{1,2,3,4,9,10,11,12\} \mid \{5,6,7,8,13,14,15\}$	$\{5,6,7,8,9,10,11,12\}$	3
5	16	$\{1,2,3,4,5,6,7,8\} \mid \{9,10,11,12,13,14,15,16\}$	$\{1,2,3,4,9,10,11,12\} \mid \{5,6,7,8,13,14,15,16\}$	$\{5,6,7,8,9,10,11,12\}$	3

Table 8: Rank-three witnesses constructed as in Appendix J (first listed class per (k,p)).

Partition-profile types

For completeness we list the multiset of block sizes occurring over all classified classes. A *type* is the multiset of m_Π-values over partitions with $m \geq 3$ blocks of circuits (partitions with two blocks have $m = 2$ and appear in every profile).

k	type	number of classes
3	$[1^3]$	1
4	$[1^7]$	2
4	$[1^4\,2^1]$	1
5	$[1^{15}]$	7
5	$[1^{12}\,2^1]$	3
5	$[1^8\,3^1]$	1
5	$[1^9\,2^2]$	1
6	$[1^{31}]$	134
6	$[1^{28}\,2^1]$	59
6	$[1^{25}\,2^2]$	12
6	$[1^{24}\,3^1]$	4
6	$[1^{22}\,2^3]$	3
6	$[1^{16}\,2^5]$	1
6	$[1^{16}\,4^1]$	1
6	$[1^{21}\,2^1\,3^1]$	1

Table 9: Types occurring among the classified minimal classes ($k \leq 6$), as computed by `scripts/gen_tables.py` from the exhaustive and source-reported classifications. The rows with $k = 6$ rely on the source-reported classification (certificates replayed with the independent engine).

K Classification methods and certificates

$k \leq 5$: independent exhaustive enumeration. `scripts/classify.py` enumerates all subsets of the affine space, tests minimality with the engine `geo2.py` (circuits by brute force from the definition, E1 criterion, single-shortening test), and groups subsets into orbits. The output (`logs/classify.json`) matches the release counts $1, 3, 12$.

$k = 6$: source-reported. The enumeration over 2^{32} subsets is not repeated here. The release GEO-I v010 (`BINARY6.md`) reports 215 classes and supplies certificates `C6_rank6.json` with the checker `checker10.py`. In this work the checker was run (output in `logs/source_checkers.txt`) and the 215 representatives were re-examined with the independent engine: each is spanning, non-embedded, with all shortenings embedded, and satisfies the counting and Parseval identities (`scripts/verify6.py`, `logs/verify6.json`). Pairwise inequivalence of the 215 representatives was checked by invariants (`scripts/inv6*.py`) together with an equivalence search (`scripts/equiv6.py`). *Completeness* of the list rests on the source and is not reproved here; nothing in the universal proof depends on it.

$p \leq 8$. Proposition 16.2 is a complete finite verification using the simplex normal form; its details are in `scripts/small_p.py` and `logs/small_p.json`.

L Reproducibility

Pinned sources. All citations to the GEO-I releases v001–v012 refer to the archived research-release repository at commit `343f9aef2b6697466058165c5fc2ddf0f00e882b`. The repository itself could not be fetched during the preparation of this manuscript; local copies of the release files supplied by the author were used, and the commit identifier was supplied by the author. These releases are archived research releases, not peer-reviewed publications.

File tree.

`tex/`	LaTeX sources (`main.tex`, `part1-5.tex`, `appendices.tex`, `refs.tex`, generated `gen_data.tex`) and `figures/`.
`scripts/`	`geo2.py` (independent engine), `classify.py`, `crosscheck.py`, `lit_checks.py`, `verify6.py`, `small_p.py`, `inv6*.py`, `equiv6.py`, `gen_tables.py`, `figs.py`.
`logs/`	JSON/text outputs of the above, including `source_checkers.txt`.
`src/`	copies of the release certificates and checkers used (`C5_rank5.json`, `C6_rank6.json`, `C9_partition.json`, `checker10.py`, `checker11.py`).

Commands. With Python 3 and the packages `numpy` and `matplotlib`, run in `scripts/`: `python3 classify.py`, `python3 verify6.py`, `python3 small_p.py`, `python3 gen_tables.py`, `python3 figs.py`; then three passes of `pdflatex main.tex` in `tex/`. The engine uses only the Python standard library (exact rational elimination with `fractions.Fraction`).

Exact versus numerical. All ranks and identities reported are exact (integer or rational arithmetic); no floating-point computation enters any claim. Floating point is used only in drawing the figures.

Environment. Python 3.13, pdfTeX 3.141592653-2.6-1.40.25 (TeX Live 2023), Linux.

M Theorem crosswalk and provenance

"Release" refers to the GEO-I archived release and file in which the statement is recorded; "Source" is the status reported there; "Here" says how the statement is established in this paper. "New" marks statements not found in the release files that were read.

Statement here	Release id	Release / file	Source status	Status here
Prop. 3.7, Thm. 4.4	B1, B2	v006 `OBSTRUCTION_RANK.md`	proved	proved in text
Thm. 5.2	G10-E1	v010 `MATHEMATICS.md` §1.1	proved	proved in text
Prop. 5.4, Cor. 5.5	–	–	–	new; proved in text; enumerated for $k \leq 5$
Prop. 6.4	G9-T5	v009 `MATHEMATICS.md` §2.4	proved+checked	proved in text
Cor. 6.5	B-NULL′	v008 `MINIMAL_RANK.md`	proved	proved in text
Prop. 6.7	B-SERIES	v008 `MINIMAL_RANK.md`	proved+checked	proved in text
Thm. 7.1	G10-F1 (closes G8-C3)	v010 `MATHEMATICS.md` §1.2	proved	proved in text
Prop. 7.2	G9-Z, G11-L5c	v009 §1; v011 §1.1	proved(+checked)	proved in text (one argument)
Thm. 8.2	G10-A1	v010 `MATHEMATICS.md` §1.3	proved	proved in text
Lem. 9.1	B-LOW (G8-T3a)	v008 `MINIMAL_RANK.md`	proved	proved in text
Thm. 9.2	G10-L5, B-COMMON	v010 §1.5; v008	proved	proved in text
Thm. 9.4	G10-C1′	v010 `MATHEMATICS.md` §1.5	proved	proved in text
Thm. 10.1	G10-R	v010 `MATHEMATICS.md` §1.4	proved	proved in text
Prop. 10.2	–	–	–	new; proved in text
Thm. 11.3, 12.1	G11-L5	v011 `MATHEMATICS.md` §1.2–1.3	proved	proved in text
Thm. 13.3	G11-L5b	v011 `MATHEMATICS.md` §1.4	proved	proved in text
Thm. 15.1	G11-L6 (sharp: G11-K1)	v011 `MATHEMATICS.md` §1.5	proved+checked	proved in text; equality case analysis new
Prop. 15.2	G8-T3c	v008 `MINIMAL_RANK.md`	proved, checked $k \leq 5$	proved in text
Prop. 16.2	(G11-T3 part)	v011 `MATHEMATICS.md` §1.7	proved+checked	finite enumeration, own code

Statement here	Release id	Release / file	Source status	Status here
Thm. 16.3	G11-T3 (=G10-C1=G11-C2)	v011 `MATHEMATICS.md` §1.7	proved+checked	text for $p \geq 9$; enumeration for $p \leq 8$
Thm. 17.1	G11-T4	v011 `RANK_UNIVERSAL.md`	proved	proved in text modulo Prop. 16.2
Cor. 17.2	G11-C3 (G10-C2 false)	v011 `RANK_UNIVERSAL.md`	false (G10-C2)	proved in text
Thm. 19.1	G8-T3b, G9-T4	v008; v009 §2.2	proved+checked	re-enumerated exhaustively
Thm. 19.2	G10-T4	v010 `BINARY6.md`	proved+checked	source-reported; representatives re-checked
Prop. 21.4	–	–	–	new; proved in text; verified on all 231 classes
Cor. 20.1	–	–	–	new; certificate only, not an algorithm
Rem. 21.2	–	Cooper–Okur, arXiv:2504.08083, Cor. 3.1	published theorem (not a release)	translation to cycle codes derived here; consistency-checked
Prop. 21.5	–	Heden's tail theorem as quoted in Kurz, arXiv:1708.01113	published theorem (not reproved)	derived from the cited theorem; not used in any proof

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